Expected Inner Work of Inexact Proximal Point Methods

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Abstract

We consider a monotone inclusion problem in a Hilbert space. Under mild assumptions, it is known that the proximal point method converges to a zero of the operator, even with inexact computations of the resolvent. We extend this analysis to get an explicit $O(1/k)$ nonasymptotic convergence rate of the squared fixed point residual and an asymptotic $o(1/k)$ convergence rate, as well as a linear convergence rate in the presence of an error bound. Assuming access to a linearly convergent iterative procedure to compute the resolvent, we show that the method requires $O(\varepsilon^{-2})$ inner steps to find a point with $\mathbf{dist}(0, Au) \leq \varepsilon$ for a general monotone inclusion problem. We validate our theoretical findings on numerical experiments and show that many popular algorithms are special cases of our framework and benefit from inexact computations.

1 Introduction

In this paper we consider the problem of a maximal monotone inclusion on a real Hilbert space $\mathcal{H}$:

$$\text{find } x \text{ such that } 0 \in Ax, \tag{1}$$

where $A : \mathcal{H} \rightrightarrows \mathcal{H}$ is a maximal monotone operator. Any convex optimization problem can be viewed as a monotone inclusion problem (1) by examining the first-order optimality condition. More broadly, convex-concave minimax problems admit a monotone inclusion formulation. Maximal monotone inclusion is a strictly larger class of problems than finding zeros of subgradients, as there are monotone operators which are not the subgradient of any convex function (**??** shows one such example). These problems arise in a variety of fields, such as image processing [Figueiredo and Nowak, 2003], communications [Schizas and Giannakis, 2006], PDEs [Douglas and Rachford, 1956], fluid dynamics [Glowinski et al., 1994], and finance [Richard and Roncalli, 2019]. As an example, we consider the conic optimization problem

$$\begin{array}{ll} \text{minimize} & f(x) + \mathcal{I}_{\mathcal{C}}(z) \\ \text{subject to} & Ex = z, \end{array} \tag{2}$$

where $x \in \mathbb{R}^n$ and $z \in \mathbb{R}^m$ are the optimization variables, $E \in \mathbb{R}^{m \times n}$, f is a smooth objective function, and $\mathcal{C}$ is a convex cone. This can be rewritten in a monotone inclusion form (1) by considering the operator

$$A_E = \begin{bmatrix} \frac{1}{c}\partial f & E^\top & -E^\top \\ -E & \frac{1}{c}\mathcal{N}_{\mathcal{C}} & I \\ E & -I & 0 \end{bmatrix}$$

where $\mathcal{N}_{\mathcal{C}}$ is the normal cone operator of $\mathcal{C}$. The operator A_E encodes the KKT conditions of the conic optimization problem (2).

A popular method for solving the monotone inclusion (1) is the proximal point method (PPM): for a stepsize γ, each iteration solves the backward inclusion $x_k \in (I + \gamma A)x_{k+1}$. However, solving the inclusion subproblem in these algorithms can be difficult and makes each iteration expensive. Rockafellar [1976a] showed that if the sum of all approximation errors is bounded, then PPM still converges asymptotically to a solution of the problem (1). This type of criterion is known as an *absolute* error condition, since it controls the absolute error between the estimated and the true subproblem solution. However, there are several practical issues with such a condition. First, checking the error condition requires knowledge of the true update, which is generally not computable. Second, the choice of such a sequence adds an infinite number of parameters for the user to tune, making practical implementations difficult. Moreover, the convergence analysis does not provide meaningful guidance on the design of such a sequence, nor does it provide guidance on how the degree of error affects the convergence of the algorithm and the complexity of intermediate computation.

To obviate this difficulty, we would like the error condition to have two properties. First, it should be easy to compute, and not require knowledge of the true update. Second, it should not depend on an external sequence of errors, but rather be a function of the current state of the algorithm in some way: a *relative* error condition.

Relaxing the inclusion to allow for error in solving the subproblem makes these algorithms much more efficient in practice. However, the inexactness in solving the subproblem reduces the effectiveness of each iteration of PPM. Thus, there is a tradeoff between exactness and convergence speed (as measured by outer iterations). It is not clear how to tune the parameters of the outer algorithm and the subproblem procedure to balance this tradeoff.

1.1 Contribution

We introduce a unified framework to analyze the per-inner iteration work of inexact PPM and provide guidelines on selecting algorithm parameters to balance the inner and outer iteration complexity. Our setting is general enough to include many popular classes of operating splitting algorithms, such as DRS, DYS, and others. We also allow for any method of solving the inner subproblem, as long as it has linear convergence in expectation. We provide an explicit bound on the total inner iterations required to find a point with $\mathbf{dist}(0, Au) \leq \delta$ in both the deterministic and stochastic cases. Additionally, we provide high-probability bounds in the randomized case for the total complexity. By incorporating warm-starts from previous iterations into the current iteration, we improve the naive worst-case bound of $O(\delta^{-2}\log(1/\delta))$ bound for total inner iteration complexity to $O(\delta^{-2})$ for a general monotone inclusion problem (1).

Our contributions are as follows:

1. We construct a general framework for analyzing degenerate preconditioned proximal point methods and provide bounds on the error at stopping (Proposition 2.1). We show an explicit convergence rate as well as linear convergence with an error bound (Theorems 3.1 and 3.2).

2. We analyze the total inner work done over the course of the algorithm and show that the method requires $O(\varepsilon^{-2})$ inner steps to find a point with $\mathbf{dist}(0, Au) \leq \varepsilon$ for a general monotone inclusion problem (Theorem 4.2). We also provide high-probability bounds on the total inner work required for such an algorithm (Theorem 4.3).

3. We analyze our results on some model choices of algorithms, such as PPM and ADMM, and infer some principled guidelines on selecting algorithm parameters (Propositions 5.1 to 5.3).

4. **TODO:** update when done We compare our theoretical results against empirical studies and find that the theorems are often conservative. In fact, most instances benefited from decreased accuracy (**??**).

1.2 Prior Work

Operator Splitting Methods. Martinet [1970] originally proposed PPM for solving variational inequalities, and Rockafellar [1976a,b] greatly advanced theoretical understanding of the method, as well as its relation to other optimization algorithms. Eckstein and Bertsekas [1992] showed that DRS is an instance of PPM applied to a certain splitting operator, showing the connection between PPM and operator splitting algorithms. Operator splitting finds many applications in convex optimization, with notable success stories such as the iterative shrinkage-thresholding algorithm (ISTA), its accelerated variant (FISTA), and the primal-dual hybrid gradient method for linear programming (PDLP) and its variants [Figueiredo and Nowak, 2003; Beck and Teboulle, 2009; Applegate et al., 2021]. Liao et al. [2026] show that under sufficient growth conditions, PPM can even be used for nonconvex optimization as well.

Inexact PPM. One of the challenges in practical implementations of PPM is to solve the backward inclusion subproblem. For example, in the convex optimization setting, this takes the form of solving a strongly convex subproblem. Although first-order methods have theoretically strong guarantees on these problems, solving the problem exactly, or even to high precision, is undesirable. Rockafellar [1976a] showed that it suffices to compute a sequence of iterates whose errors form a summable sequence. Eckstein and Yao [2018] provide an affirmative answer to the question of designing a relative error criterion for Douglas–Rachford and the alternating direction method of multipliers (ADMM), and their result was extended by Svaiter [2019] and Barré et al. [2023].

More recently, Bredies et al. [2022] showed that many existing operator splitting methods, such as Chambolle–Pock, Douglas–Rachford, Peaceman–Rachford, and Davis–Yin, are instances of *degenerate preconditioned* proximal point methods. That is, we precondition the monotone inclusion problem (1) by replacing the scalar stepsize γ with a matrix $M \succeq 0$ and evaluating $(I + M^{-1}A)^{-1} = (M + A)^{-1}M$. Under mild assumptions, even when M is not invertible, the operator $(M + A)^{-1}M$ still exists and can be computed. These ideas can be generalized to relatively inexact resolvent evaluations with a broader class of preconditioners, such as in Alves et al. [2026].

Integrated Complexity Analysis Analyzing the total complexity of the method, which would include the overall number of inner iterations required to solve the inclusion problem, is more delicate. Machart et al. [2012] consider the complexity of inexact proximal gradient methods, but they only consider the case of absolute errors, with the standard summability conditions. Moreover, they only consider a specific instance of PPM, whereas our setting is general enough to cover many more operator splitting methods. Lin et al. [2018] consider a model of global complexity, factoring in inner and outer complexity, but their error criterion is not generally computable, requiring knowledge of the exact solution to the subproblem or access to the dual function. Alves and Geremia [2019, Theorems 4.3 and 4.4] provide analysis in a similar setting but only show asymptotic guarantees of inner complexity. Moreover, they do not consider the effect of warm-starting, resulting in an overall complexity of $O(\delta^{-2}\log(1/\delta))$, compared to our $O(\delta-2)$ result. Li and Qu [2021] consider similar types of inner procedures, but they focus on one particular instance of the augmented Lagrangian method and show worse total complexity guarantees. Du et al. [2026] design an inexact Newton method utilizing sketch-and-project methods and show complexity guarantees for inexact second-order methods.

2 The Degenerate Preconditioned Proximal Point Method

We make the following assumption to ensure that all the operations we take in this paper are well-defined.

Assumption 2.1. *We assume that A is a maximal monotone operator $A : \mathcal{H} \rightrightarrows \mathcal{H}$ on the Hilbert space $\mathcal{H}$ with at least one zero:* $\mathbf{zer}\, A \neq \varnothing$. *We assume that the map $C : \mathcal{D} \to \mathcal{H}$ is a bounded linear map between Hilbert spaces, that C^* is surjective, and that $(CC^* + \gamma A)^{-1}CC^*$ is defined everywhere.*

Definition 2.1. Define the operator B by its graph as

$$\mathbf{gra}\, B = \{(C^*u, z) \mid Cz \in Au\}.$$

Since C^* is surjective (Assumption 2.1), the resolvent $J := (I + \gamma B)^{-1}$ is defined on the entire space $\mathcal{D}$ (Lemma 2.1). Note that B itself need not have full domain. In particular, $\mathbf{dom}\, B \subseteq C^* \mathbf{dom}\, A$, with strict inclusion a possibility. One way of viewing this definition is that B is a projection of A onto a lower-dimensional space. Proposition 5.3 shows that this perspective enables us to view the iterates of ADMM as the projections of a higher-dimensional PPM.

Lemma 2.1. *For B defined as in Definition 2.1, under Assumption 2.1, the following hold:*

1. $\mathbf{zer}\, B = C^* \mathbf{zer}\, A$.
2. *For $x \in \mathcal{D}$, $w = (I+\gamma B)^{-1}x$ if and only if there exist $v \in \mathcal{H}$ and $z \in \mathcal{D}$ such that $x = C^*v + \gamma z$, $Cz \in Av$, and $w = C^*v$. Moreover, such v and z exist for every $x \in \mathcal{D}$.*
3. *B is maximal monotone.*

Proof. The first claim is immediate. The second follows from Assumption 2.1 and manipulating the definition of B:

$$\begin{aligned}(I + \gamma B)^{-1}x = w &\iff x \in (I + \gamma B)w \\ &\iff x \in (I + \gamma B)w,\ w = C^*v \text{ for some } v \text{ with } C^*v \in \mathbf{dom}\, B \\ &\iff x = C^*v + \gamma z,\ Cz \in Av,\ w = C^*v.\end{aligned}$$

For existence, fix $x \in \mathcal{D}$ and, since C^* is surjective, we can write $x = C^*\tilde{x}$. By Assumption 2.1 there exists v with $CC^*\tilde{x} \in (CC^* + \gamma A)v$, i.e., $\gamma^{-1}CC^*(\tilde{x} - v) \in Av$. Setting $z = \gamma^{-1}C^*(\tilde{x} - v)$ gives $Cz \in Av$ and $C^*v + \gamma z = C^*\tilde{x} = x$.

The third item follows from the fact that

$$\langle C^*u_1 - C^*u_2, z_1 - z_2\rangle = \langle u_1 - u_2, Cz_1 - Cz_2\rangle \geq 0,$$

since $Cz_i \in Au_i$ and A is monotone. Maximality follows from Assumption 2.1: by item 2, $\mathcal{R}(I+\gamma B) = \mathcal{D}$, so B is maximal by Minty's theorem [Bauschke and Combettes, 2017, Theorem 21.1]. □

In the case where A is not necessarily monotone, it may still hold that B is maximal monotone. We make this assumption explicit.

Assumption 2.2. *We assume that B defined in Definition 2.1 is a maximal monotone operator with at least one zero:* $\mathbf{zer}\, B \neq \varnothing$. *We also assume that the map $C : \mathcal{D} \to \mathcal{H}$ is a bounded linear map between Hilbert spaces with C^* surjective.*

Algorithm 1 Inexact Proximal Point Method

1: **Initialize:** $w_0 \in \mathcal{D}$, inexactness $\sigma \in [0, 1)$, stepsize $\gamma > 0$, stride $s \in \mathbb{N}$, and stopping tolerance δ
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Produce a pair $(u_k, z_k) \in \mathcal{H} \times \mathcal{D}$ with

$$Cz_k \in Au_k, \qquad \gamma z_k + C^* u_k \approx w_k.$$

4: **if** $\|z_k\| \leq \delta$ **then**
5: Stop and return (z_k, u_k, w_k).
6: **else if** $\|\gamma z_k + C^* u_k - w_k\| \leq \sigma \|C^* u_k - w_k\|$ **then**
7: Set $w_{k+1} = w_k - \gamma z_k$.
8: **else**
9: Improve the pair $(u_k, z_k) \in \mathcal{H} \times \mathcal{D}$ with $Cz_k \in Au_k$ via the iterative procedure s times and go to line 4.
10: **end if**
11: **end for**

Note that Assumption 2.1 is strictly stronger than Assumption 2.2. Proposition 5.4 shows an example where Assumption 2.2 holds but Assumption 2.1 fails.

Using these assumptions, we can now introduce Algorithm 1, whose convergence properties we will analyze in this section. The following proposition shows that at stopping, the returned iterates provide a certificate of distance to optimality.

Proposition 2.1. *Every output from the $\|z\| \leq \delta$ terminal test of Algorithm 1 satisfies*

$$Cz \in Au, \qquad \mathbf{dist}(0, Au) \leq \|Cz\| \leq \|C\|\delta.$$

Moreover, after stopping we also have

$$\|C^* u - JC^* u\| \leq \gamma\delta.$$

That is, the fixed point residual also satisfies the bound at halting. Note that the returned iterate may not satisfy the relative error criterion.

Proof. The first claim is immediate from the definition of Algorithm 1. For the second, let $q = C^* u - JC^* u = (I - J)C^* u$. Then,

$$\left(C^* u - q, \frac{q}{\gamma}\right) = \left(JC^* u, \frac{q}{\gamma}\right) = \left(JC^* u, \frac{(I - J)C^* u}{\gamma}\right).$$

Using Assumption 2.2, there exists x such that $C^* x = JC^* u$. Then,

$$\begin{aligned} C^* x = (I + \gamma B)^{-1} C^* u &\iff C^* u \in C^* x + \gamma B C^* x \\ &\iff C^* u = C^* x + \gamma\xi, \quad C\xi \in Ax \\ &\iff \frac{C^*(u - x)}{\gamma} = \xi, \quad C\xi \in Ax. \end{aligned}$$

Thus, we can write

$$\left(JC^*u, \frac{(I-J)C^*u}{\gamma}\right) = (C^*x, \gamma^{-1}C^*(u-x)), \quad \frac{C^*(u-x)}{\gamma} = \xi, C\xi \in Ax$$
$$= (C^*x, \xi), \quad \frac{C^*(u-x)}{\gamma} = \xi, C\xi \in Ax,$$

so $\left(C^*u - q, \frac{q}{\gamma}\right) \in \mathbf{gra}\, B$. Since $(C^*u, z) \in \mathbf{gra}\, B$ and B is monotone,

$$0 \leq \left\langle (C^*u - q) - C^*u, \frac{q}{\gamma} - z \right\rangle = \left\langle q, z - \frac{q}{\gamma} \right\rangle.$$

Thus, $\langle q, \gamma z - q\rangle \geq 0$ and hence

$$\|\gamma z\|^2 = \|\gamma z - q\|^2 + 2\langle \gamma z - q, q\rangle + \|q\|^2 \geq \|q\|^2.$$

Applying the stopping rule $\|z_k\| \leq \delta$ from Algorithm 1 (line 4) completes the proof. □

3 Outer Convergence Guarantees

In this section, we present a quantitative estimate on the convergence of Algorithm 1. The inexactness affects convergence only through a multiplicative factor, which allows the rest of the analysis to follow. The next lemma is needed to show the convergence of Algorithm 1.

Lemma 3.1. *The resolvent J and its fixed-point residual $R := I - J$ are firmly nonexpansive. If $(y, z) \in \mathbf{gra}\, B$ is a candidate update pair (as in line 3 of Algorithm 1 with $y = C^*u$) at w, then*

$$y = J(\gamma z + y), \tag{3}$$
$$\gamma z = R(\gamma z + y), \tag{4}$$
$$\|y - J(w)\|^2 + \|\gamma z - R(w)\|^2 \leq \|\gamma z + y - w\|^2. \tag{5}$$

If the pair (y, z) passes the relative error test, then we also have

$$(1-\sigma)\|y - w\| \leq \|R(w)\| \leq (1+\sigma)\|y - w\|, \tag{6}$$
$$(1-\sigma)\|y - w\| \leq \|\gamma z\| \leq (1+\sigma)\|y - w\|, \tag{7}$$
$$\|\gamma z + y - w\| \leq \frac{\sigma}{1-\sigma}\|R(w)\|, \tag{8}$$
$$\|\gamma z\| \leq \frac{\|R(w)\|}{1-\sigma}. \tag{9}$$

Moreover, $\|R(w)\| \leq \mathbf{dist}\,(w, \mathbf{zer}\, B)$.

Proof. For the first equation, note that $y = C^*u$ and thus

$$y = J(\gamma z + y) \iff \gamma z + y \in y + \gamma By \iff z \in BC^*u,$$

which is the definition of B. Next,

$$R(\gamma z + y) = \gamma z + y - J(\gamma z + y) = \gamma z + y - y = \gamma z.$$

Firm nonexpansiveness follows from Assumption 2.2 and the firm nonexpansiveness of resolvents of maximal monotone operators [Bauschke and Combettes, 2017, Corollary 23.11].

Next, by the monotonicity of B,

$$\langle y_1 - y_2, z_1 - z_2 \rangle \geq 0,$$

so $\langle y_1 - y_2, \gamma z_1 - \gamma z_2 \rangle \geq 0$ as well. For an exact pair, $w = y + \gamma z$ with $z \in By$,

$$\begin{aligned} \|w_1 - w_2\|^2 &= \|\gamma z_1 - \gamma z_2\|^2 + \|y_1 - y_2\|^2 + 2\langle y_1 - y_2, \gamma z_1 - \gamma z_2 \rangle \\ &\geq \|\gamma z_1 - \gamma z_2\|^2 + \|y_1 - y_2\|^2. \end{aligned} \tag{10}$$

Note that since $w = y + \gamma z$ with $z \in By$, we have $y = J(w)$ and $\gamma z = R(w)$. That is, (10) states $\|w_1 - w_2\|^2 \geq \|R(w_1) - R(w_2)\|^2 + \|J(w_1) - J(w_2)\|^2$ for all w_1 and w_2.

Let $e = \gamma z + y - w$, the graph residual. Note that $w + e = \gamma z + y$. Since $y = J(\gamma z + y) = J(w + e)$,

$$\gamma z = w + e - y = w + e - J(w + e) = (I - J)(w + e) = R(w + e).$$

Thus, using the resolvent identities $y = J(\gamma z + y)$ and $\gamma z = R(\gamma z + y)$ (3) and (4) and the component bound $\|w_1 - w_2\|^2 \geq \|\gamma z_1 - \gamma z_2\|^2 + \|y_1 - y_2\|^2$ (10) with $w_1 = w + e$ and $w_2 = w$,

$$\begin{aligned} \|e\|^2 = \|(w + e) - w\|^2 &\geq \|R(w + e) - R(w)\|^2 + \|J(w + e) - J(w)\|^2 \\ &= \|R(\gamma z + y) - R(w)\|^2 + \|J(\gamma z + y) - J(w)\|^2 \\ &= \|\gamma z - R(w)\|^2 + \|y - J(w)\|^2. \end{aligned}$$

Now suppose the relative error test $\|e\| \leq \sigma \|y - w\|$ holds. By the reverse triangle inequality,

$$|\|y - w\| - \|R(w)\|| = |\|y - w\| - \|w - J(w)\|| \leq \|y - J(w)\| \leq \|e\| \leq \sigma \|y - w\|.$$

This is the residual bound (6). Next, $\gamma z = e - (y - w)$, so the triangle inequality gives

$$(1 - \sigma)\|y - w\| \leq \|y - w\| - \|e\| \leq \|\gamma z\| \leq \|y - w\| + \|e\| \leq (1 + \sigma)\|y - w\|,$$

which is the update bound (7). Combining $\|e\| \leq \sigma \|y - w\|$ with the left inequality of (6) gives the error bound (8). Finally, $\|\gamma z - R(w)\| \leq \|e\|$ and $\|e\| \leq \frac{\sigma}{1-\sigma}\|R(w)\|$ by the component bound (5) and the error bound (8), so

$$\|\gamma z\| \leq \|R(w)\| + \|\gamma z - R(w)\| \leq \|R(w)\| + \|e\| \leq \left(1 + \frac{\sigma}{1 - \sigma}\right) \|R(w)\| = \frac{\|R(w)\|}{1 - \sigma},$$

which is the bound (9).

Lastly, if $w^\star \in \mathbf{zer}\, B$, then $Jw^\star = w^\star$. Thus, $R(w^\star) = 0$, so by the nonexpansiveness of R,

$$\|R(w)\| = \|R(w) - R(w^\star)\| \leq \|w - w^\star\|.$$

Taking infimums shows the desired result. □

Similar results to Proposition 2.1 and Lemma 3.1 can be found in Rockafellar [1976b, Propositions 1, 2, and 3], Bredies et al. [2022, Lemma 2.6], and Alves et al. [2026, Proposition 2.3] (among others), but we present additional identities and estimates here for our purposes.

Lemma 3.2. *For an accepted pair with $w^+ = w - \gamma z$ and any $w^\star \in \mathbf{zer}\, B$,*

$$\|w^+ - w^\star\|^2 \leq \|w - w^\star\|^2 - \|y - w\|^2 + \|\gamma z + y - w\|^2.$$

Under the relative error rule, this yields

$$\|w^+ - w^\star\|^2 \leq \|w - w^\star\|^2 - \frac{\gamma^2(1-\sigma)}{1+\sigma}\|z\|^2,$$
$$\|w^+ - w^\star\|^2 \leq \|w - w^\star\|^2 - \frac{1-\sigma}{1+\sigma}\|R(w)\|^2.$$

Proof. Recall that we defined $e = \gamma z + y - w$, the graph residual. The claim then follows immediately from expanding $w = \gamma z + y - e$:

$$\begin{aligned}\|w - w^\star\|^2 &= \|w - \gamma z - w^\star\|^2 + 2\langle w - \gamma z - w^\star, \gamma z\rangle + \|\gamma z\|^2 \\ &= \|w - \gamma z - w^\star\|^2 + 2\langle y - e - w^\star, \gamma z\rangle + \|\gamma z\|^2 \\ &= \|w - \gamma z - w^\star\|^2 + 2\langle y - w^\star, \gamma z\rangle - 2\langle e, \gamma z\rangle + \|\gamma z\|^2.\end{aligned}$$

Using the monotonicity of B (with $(w^\star, 0), (y, z) \in \mathbf{gra}\, B$) on the first inner product term and then completing the square yields

$$\begin{aligned}\|w - w^\star\|^2 &\geq \|w - \gamma z - w^\star\|^2 - \|e\|^2 + \|e - \gamma z\|^2 \\ &= \|w - \gamma z - w^\star\|^2 - \|\gamma z + y - w\|^2 + \|w - y\|^2.\end{aligned}$$

Applying the relative error rule, $\|e\| \leq \sigma\|y - w\|$, and Lemma 3.1 yields

$$\begin{aligned}\|w - \gamma z - w^\star\|^2 &\leq \|w - w^\star\|^2 + \|\gamma z + y - w\|^2 - \|w - y\|^2 \\ &\leq \|w - w^\star\|^2 + \sigma^2\|y - w\|^2 - \|w - y\|^2 \\ &= \|w - w^\star\|^2 - (1 - \sigma^2)\|w - y\|^2 \\ &\leq \|w - w^\star\|^2 - \frac{(1-\sigma^2)}{(1+\sigma)^2}\|\gamma z\|^2.\end{aligned}$$

The last inequality uses $\|\gamma z\| \leq (1+\sigma)\|y - w\|$ from the update bound (7) of Lemma 3.1, which yields the first bound. Using $\|R(w)\| \leq (1+\sigma)\|y - w\|$ from the residual bound (6) of Lemma 3.1 instead gives the second. □

Using Lemma 3.2, we immediately obtain the following finite termination guarantee on Algorithm 1. We remark that this matches well-known results on the $o(k^{-1})$ convergence rate of the squared fixed point residual.

Theorem 3.1. *After K accepted updates,*

$$\sum_{k=0}^{K-1}\|\gamma z_k\|^2 \leq \frac{(1+\sigma)\,\mathbf{dist}\,(w_0, \mathbf{zer}\, B)^2}{1-\sigma}.$$

As a result,

$$\min_{k=0,\ldots,K-1}\|Cz_k\| \leq \frac{\|C\|\sqrt{1+\sigma}\,\mathbf{dist}\,(w_0, \mathbf{zer}\, B)}{\gamma\sqrt{(1-\sigma)K}}.$$

Moreover, if K_δ is the number of accepted updates before returning a point satisfying $\|z_{K_\delta}\| \leq \delta$ (the terminal test of Algorithm 1), then

$$K_\delta \leq \left\lfloor \frac{(1+\sigma)\,\mathbf{dist}\,(w_0, \mathbf{zer}\,B)^2}{\gamma^2\delta^2(1-\sigma)} \right\rfloor =: K_\delta^+.$$

Moreover, if $K_\delta > 0$, then $K_\delta < K_\delta^+$.

Proof. The first two claims follow immediately from the telescoping identity of Lemma 3.2 and Proposition 2.1. The last claim follows from the fact that for all iterations before K_δ, the accepted iterate satisfies $\|z_k\| > \delta$. Therefore, applying the first claim with $K = K_\delta$,

$$\frac{(1+\sigma)\,\mathbf{dist}\,(w_0, \mathbf{zer}\,B)^2}{1-\sigma} \geq \sum_{k=0}^{K_\delta - 1} \|\gamma z_k\|^2 \geq K_\delta \gamma^2 \delta^2,$$

where the second inequality is strict if $K_\delta \geq 1$ and trivial if $K_\delta = 0$. Recognizing that K_δ is an integer and rearranging yields the result. □

Corollary 3.1.1. *Using Theorem 3.1 and setting $\delta = 0$, we find $\min_{k=0,\ldots,K-1}\|z_k\|^2 = o(K^{-1})$.*

Proof. From the first part of Theorem 3.1, we know that $\sum_{k=0}^{\infty}\|\gamma z_k\|^2 \leq \frac{(1+\sigma)\,\mathbf{dist}\,(w_0,\mathbf{zer}\,B)^2}{1-\sigma}$. Thus,

$$\frac{K}{2} \min_{k=0,\ldots,K-1}\|\gamma z_k\|^2 \leq \sum_{k=\lfloor \frac{K}{2} \rfloor}^{K-1} \|\gamma z_k\|^2 \leq \sum_{k=\lfloor \frac{K}{2} \rfloor}^{\infty} \|\gamma z_k\|^2 \to 0$$

as $K \to \infty$. Since $\gamma > 0$, we have the desired asymptotic convergence rate. □

Remark 3.1. By Lemma 3.1, bounds on $\|\gamma z_k\|$ can be converted into bounds on $\|R(w)\|$. For brevity, we omit them here.

In essence, Theorem 3.1 shows how many iterations are required before stopping, and Proposition 2.1 guarantees the accuracy of the iterate at halting. We now show that under an error bound guarantee, we obtain linear convergence of the method, where the contraction factor is attenuated by the inexactness of the method.

Theorem 3.2. *Under an error bound guarantee of the form*

$$\mathbf{dist}(w_k, \mathbf{zer}\,B) \leq \Lambda \|R(w_k)\|$$

for all k, where $\Lambda \geq 1$, we find that

$$\mathbf{dist}(w_k, \mathbf{zer}\,B)^2 \leq \left(1 - \frac{1-\sigma}{\Lambda^2(1+\sigma)}\right)^k \mathbf{dist}(w_0, \mathbf{zer}\,B)^2.$$

Moreover, the stopping time $K_\delta = \min\{k \mid \|z_k\| \leq \delta\}$ obeys

$$K_\delta \leq \frac{\log\left[\mathbf{dist}(w_0, \mathbf{zer}\,B)^2\right] - \log\left[(1-\sigma)^2\gamma^2\delta^2\right]}{\log\frac{1}{\omega}},$$

where $\omega = 1 - \frac{1-\sigma}{\Lambda^2(1+\sigma)}$.

Proof. From Lemma 3.2,

$$\begin{aligned}\mathbf{dist}(w^+, \mathbf{zer}\, B)^2 &\leq \mathbf{dist}(w, \mathbf{zer}\, B)^2 - \frac{1-\sigma}{1+\sigma}\|R(w)\|^2 \\ &\leq \left(1 - \frac{1-\sigma}{\Lambda^2(1+\sigma)}\right)\mathbf{dist}(w, \mathbf{zer}\, B)^2.\end{aligned}$$

Unraveling the recursion completes the proof of the first inequality.

Next, by the bound (9) of Lemma 3.1, $\|z_k\| \leq \frac{\mathbf{dist}(w_k, \mathbf{zer}\, B)}{\gamma(1-\sigma)}$ for updates passing the relative error test. Thus, a sufficient condition for stopping is $\mathbf{dist}(w_k, \mathbf{zer}\, B) \leq (1-\sigma)\gamma\delta$, which happens after at most

$$\frac{\log\left[\mathbf{dist}(w_0, \mathbf{zer}\, B)^2\right] - \log\left[(1-\sigma)^2\gamma^2\delta^2\right]}{\log\frac{1}{\omega}}$$

accepted updates, with $\omega = 1 - \frac{1-\sigma}{\Lambda^2(1+\sigma)}$. □

Remark 3.2. The convergence results of Theorems 3.1 and 3.2 can be modified to cover the case of deterministic time-varying stepsizes as well. This case is a simple application of Lemma 3.2, and we omit it here.

4 Inner Complexity Properties

The analysis of Section 3 ignores the fact that solving the proximal subproblem is usually not trivial. For example, the proximal operation on the quadratic function $f(x) = \frac{1}{2}x^\top Qx + d^\top x$ requires solving the system $(\gamma Q + I)x^+ = x - \gamma d$. This can be nontrivial in general, and it would be ideal to obtain guarantees for the complexity of the algorithm in terms of the number of inner iterations, as well as to connect this to the general structure of the problem and parameters.

We begin with some definitions to establish the setting of the problem.

Definition 4.1. Let $\mathbf{P}_k(\mathcal{E}) := \mathbf{P}(\mathcal{E} \mid \mathcal{F}_k)$, and $\mathbf{E}_k$ similarly. Here $\{\mathcal{F}_k\}$ represents the filtration generated by the randomness of the algorithm up to the start of *outer* iteration k, i.e., all inner and outer history before the kth inner solve begins. In particular w_k, $R(w_k)$, the warm start of the kth inner solve, and the event $\{K_\delta \geq k\}$ are $\mathcal{F}_k$-measurable. We write N_k for the number of inner iterations performed during outer iteration k.

Assumption 4.1. *We assume that the inner procedure referred to in Algorithm 1 has a guarantee of the form*

$$\mathbf{E}_k[\|\gamma z^l + y^l - w_k\|^2] \leq C_0\alpha^l\mathbf{E}_k[\|\gamma z^0 + y^0 - w_k\|^2],$$

where $0 < \alpha < 1$ *and* $C_0 \geq 1$ *are constants that do not depend on* k *or* l*. Here the superscripts represent* inner *iterations (those in the loop of lines 4–10 in Algorithm 1).*

This may seem like a strong assumption, since it asks for an inner procedure whose residual contracts linearly.[1] It is, however, mild. When $A = \partial f$ for a closed, proper, and convex function f, the resolvent is the proximal operator of f. Thus, the subproblem $\min_x f(x) + \frac{1}{2\gamma}\|x - w_k\|^2$ is $\frac{1}{\gamma}$-strongly convex, so any linearly convergent method for strongly convex problems qualifies. For smooth f, gradient descent is one such inner procedure, with the constants of Proposition 5.1. In

[1]The term "linear convergence" is often referred to as "geometric convergence" outside of optimization contexts.

the quadratic example above, where $f(x) = \frac{1}{2}x^\top Qx + d^\top x$ and the subproblem is to solve the linear system $(\gamma Q + I)x^+ = x - \gamma d$, the generalized minimal residual method (GMRES) contracts the residual linearly, with the constants used in **??**. A method that does not converge linearly on a strongly convex subproblem would be a poor choice of inner solver.

Theorem 4.1. *Under Assumptions 2.2 and 4.1, if $\sigma \in (0,1)$, we have the following bounds on the conditional expectation of the inner work of Algorithm 1 at outer iteration k, for every integer $l \geq 0$:*

$$\begin{aligned}\mathbf{P}_k(N_k > sl) &\leq \min\{1, \beta_k \alpha^{sl}\},\\ \mathbf{E}_k[N_k] &\leq \frac{(\log \beta_k)_+}{\log(1/\alpha)} + \frac{s}{1-\alpha^s},\end{aligned}$$

where

$$\beta_k = \frac{(1+\sigma)^2 C_0 \mathbf{E}_k[\|\gamma z^0 + y^0 - w_k\|^2]}{\sigma^2 \max\{\|R(w_k)\|^2, \gamma^2\delta^2\}}.$$

Proof. On the event $\{N_k > sl\}$, both tests in Algorithm 1 fail at the check after sl inner iterations. Write $y = C^* u^{sl}$ and $e = \gamma z^{sl} + y - w_k$. Thus, $\|\gamma z^{sl} + C^* u^{sl} - w_k\| > \sigma \|C^* u^{sl} - w_k\|$. Note that Lemma 3.1 gives $\|y - J(w_k)\| \leq \|e\|$, so

$$\|R(w_k)\| \leq \|y - w_k\| + \|y - J(w_k)\| < \frac{\|e\|}{\sigma} + \|e\| = \frac{1+\sigma}{\sigma}\|e\|.$$

Similarly, since $\gamma z^{sl} = e - (y - w_k)$ and the terminal test fails ($\|z^{sl}\| > \delta$),

$$\gamma\delta < \|\gamma z^{sl}\| \leq \|e\| + \|y - w_k\| < \frac{1+\sigma}{\sigma}\|e\|.$$

Thus, by the conditional Markov's inequality,

$$\begin{aligned}\mathbf{P}_k(N_k > sl) &\leq \mathbf{P}_k\left(\|\gamma z^{sl} + C^* u^{sl} - w_k\| > \frac{\sigma}{1+\sigma}\max\{\|R(w_k)\|, \gamma\delta\}\right)\\ &\leq \frac{\mathbf{E}_k[\|\gamma z^{sl} + C^* u^{sl} - w_k\|^2]}{\left(\frac{\sigma}{1+\sigma}\max\{\|R(w_k)\|, \gamma\delta\}\right)^2}\\ &\leq \beta_k \alpha^{sl}.\end{aligned}$$

The expectation bound holds since N_k takes values $0, s, 2s, \ldots$ and thus

$$\begin{aligned}\mathbf{E}_k[N_k] = \sum_{l=0}^{\infty} \mathbf{P}_k(N_k > l) &\leq s \sum_{l=0}^{\infty} \min\{1, \beta_k \alpha^{sl}\}\\ &= s\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil + s\beta_k \sum_{l=\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil}^{\infty} \alpha^{sl}\\ &= s\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil + s\beta_k \frac{\alpha^{s\cdot\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil}}{1-\alpha^s}\\ &\leq \frac{(\log \beta_k)_+}{\log(1/\alpha)} + \frac{s}{1-\alpha^s}.\end{aligned}$$

For the last step, let $l^\star := \frac{(\log \beta_k)_+}{s \log(1/\alpha)}$ and note that $\lceil l^\star \rceil - l^\star \in [0,1)$. If $\beta_k \geq 1$, then $\beta_k \alpha^{s l^\star} = 1$, so $\beta_k \alpha^{s\lceil l^\star \rceil} = \alpha^{s(\lceil l^\star \rceil - l^\star)}$ and the previous line equals $s\left(l^\star + (\lceil l^\star \rceil - l^\star) + \frac{\alpha^{s(\lceil l^\star \rceil - l^\star)}}{1-\alpha^s}\right)$. The function $x \mapsto x + \frac{\alpha^{sx}}{1-\alpha^s}$ is convex on $[0,1]$ and equals $\frac{1}{1-\alpha^s}$ at both endpoints, so it is at most $\frac{1}{1-\alpha^s}$ on $[0,1)$. If $\beta_k < 1$, then $l^\star = 0$ and the sum equals $\frac{\beta_k}{1-\alpha^s} \leq \frac{1}{1-\alpha^s}$. In both cases, $\sum_{l=0}^{\infty} \min\{1, \beta_k \alpha^{sl}\} \leq l^\star + \frac{1}{1-\alpha^s}$, which is the claim. □

4.1 Warm Start Analysis

The analysis in Theorem 4.1 shows that the inner stopping time N_k has exponentially decaying tails and thus is finite almost surely. However, the structure of the guarantee in Assumption 4.1 implies that the inner method is sensitive to the choice of initialization. In fact, this is one of the motivations for using the PPM: we can solve a regularized subproblem at each iteration and warm-start the inner solver with the previous iterate. The following lemma bounds the drift of the solution due to this choice of warm-starting.

Lemma 4.1. *Suppose that the inner solve in Algorithm 1 is warm-started at the previously accepted solution. Let (u_k, z_k) be the pair accepted at outer iteration k (as in Algorithm 1), $y_k = C^* u_k$, and $e_k = \gamma z_k + y_k - w_k$, so that $w_{k+1} = w_k - \gamma z_k$. Then, we have the following bounds on the drift of the inner system:*

$$\|\gamma z_k + C^* u_k - w_{k+1}\| \leq G_\sigma \|R(w_k)\|, \tag{11a}$$

$$\|R(w_{k+1})\| \leq H_\sigma \|R(w_k)\|, \tag{11b}$$

where, with $\sigma_1 := \frac{\sqrt{13}-1}{4}$ and $\sigma_2 := \frac{\sqrt{7}-1}{2}$,

$$H_\sigma := \frac{1}{\sqrt{1-\sigma^2}}, \qquad G_\sigma := \begin{cases} 1 + 2\sigma, & 0 < \sigma \leq \sigma_1, \\ 2\sqrt{\dfrac{3(1-\sigma^2)}{3-4\sigma^2}}, & \sigma_1 \leq \sigma \leq \sigma_2, \\ \dfrac{2\sigma - 1}{1-\sigma}, & \sigma_2 \leq \sigma < 1. \end{cases} \tag{12}$$

The function $\sigma \mapsto G_\sigma$ is continuous and increasing, and $G_\sigma \geq H_\sigma$. Moreover, both constants H_σ and G_σ in (11) are tight.

The proof can be found in Appendix B.

Corollary 4.1.1. *Conditionally, for $k \geq 1$ on the event $\{K_\delta \geq k\}$,*

$$\beta_k \leq \frac{D_\sigma \|R(w_{k-1})\|^2}{\max\{\|R(w_k)\|^2, \gamma^2 \delta^2\}},$$

where

$$D_\sigma := \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}.$$

Combining Theorem 4.1 and Lemma 4.1 yields a bound on the total inner work done to reach a residual of size at most δ.

Theorem 4.2. *Suppose Assumptions 2.2 and 4.1 hold, $\sigma \in (0,1)$, and for $k \geq 1$ the inner solver is warm-started at the previously accepted pair as in Lemma 4.1. Let $D = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}$ as in Corollary 4.1.1, and let β_0 be the quantity of Theorem 4.1 for the (cold-started) first inner solve. Then*

$$\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] \leq \mathbf{E}[K_\delta]\left(\frac{s}{1-\alpha^s} + \frac{\log D_\sigma}{\log(1/\alpha)}\right) + \frac{1}{\log(1/\alpha)}\left(\frac{s\log(1/\alpha)}{1-\alpha^s} + (\log\beta_0)_+ + \log\frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2}\right). \tag{13}$$

As a consequence, $\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] = O(\delta^{-2})$.

Proof. Let $r_k := \max\{\|R(w_k)\|, \gamma\delta\}$, $c_1 := \frac{s}{1-\alpha^s}$, and

$$T_k := c_1 + \frac{(\log\beta_k)_+}{\log(1/\alpha)}.$$

By Theorem 4.1, $\mathbf{E}_k[N_k] \leq T_k$ on $\{K_\delta \geq k\}$, and T_k is $\mathcal{F}_k$-measurable.

Fix $k \geq 1$ on $\{K_\delta \geq k\}$. By Corollary 4.1.1, $\beta_k \leq \frac{D r_{k-1}^2}{r_k^2}$. By Lemma 4.1 and the fact that $H_\sigma \geq 1$, $r_k \leq H_\sigma r_{k-1}$. Thus,

$$\log D_\sigma + \log\frac{r_{k-1}^2}{r_k^2} \geq \log\frac{D}{H_\sigma^2} \geq 0,$$

since $D = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2} \geq G_\sigma^2 \geq H_\sigma^2$. Here $C_0 \geq 1$, $\frac{1+\sigma}{\sigma} > 1$, and $G_\sigma \geq H_\sigma$ by Lemma 4.1. Therefore, the maximum with 0 in the $(\log)_+$ term can be dropped:

$$(\log\beta_k)_+ \leq \max\left\{0, \log D_\sigma + \log\frac{r_{k-1}^2}{r_k^2}\right\} = \log D_\sigma + \log\frac{r_{k-1}^2}{r_k^2}.$$

Summing and telescoping, for every $K \leq K_\delta$,

$$\sum_{k=1}^{K}(\log\beta_k)_+ \leq K\log D_\sigma + \log\frac{r_0^2}{r_K^2} \leq K\log D_\sigma + \log\frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}, \tag{14}$$

using $r_K \geq \gamma\delta$ to eliminate the maximum in the denominator. Taking $K = K_\delta$ gives the almost sure bound

$$\sum_{k=0}^{K_\delta} T_k \leq (K_\delta+1)c_1 + \frac{1}{\log(1/\alpha)}\left((\log\beta_0)_+ + K_\delta\log D_\sigma + \log\frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}\right).$$

By Tonelli's theorem and the tower property,

$$\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] = \sum_{k=0}^{\infty}\mathbf{E}\left[\mathbf{1}_{\{K_\delta\geq k\}}\mathbf{E}_k[N_k]\right] \leq \sum_{k=0}^{\infty}\mathbf{E}\left[\mathbf{1}_{\{K_\delta\geq k\}}T_k\right] = \mathbf{E}\left[\sum_{k=0}^{K_\delta} T_k\right].$$

Taking expectations in the almost sure bound yields the theorem. All quantities are finite because $K_\delta \leq K_\delta^+$ almost surely (Theorem 3.1).

Lastly, by Theorem 3.1, $K_\delta \leq K_\delta^+ = O(\delta^{-2})$. Thus, $\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] = O(\delta^{-2})$. □

Remark 4.1. The $\log(\delta^{-1})$ term cannot in general be omitted from a bound in terms of the actual number of outer updates. Proposition A.1 gives a fixed smooth convex instance with $K_\delta = 1$ whose total inner work has exactly the logarithmic leading term in (13). Separately, Proposition A.2 gives a family attaining $\Theta(\delta^{-2})$ total inner work with fixed algorithm parameters and bounded initialization quantities.

The choice of warm start is flexible. The main ingredient in the proof is a bound on the initialization in terms of the previous fixed point residual and a bound on the next fixed point residual in terms of the current. That is, it suffices to find constants G and H such that

$$\begin{aligned} \mathbf{E}_{k+1}[\|\gamma z^0 + C^* u^0 - w_{k+1}\|^2] &\le G^2 \|R(w_k)\|^2, \quad \text{(warm-start bound)} \\ \|R(w_{k+1})\| &\le H \|R(w_k)\|, \quad \text{(residual drift bound)} \end{aligned}$$

where $H \ge 1$. The proof of Theorem 4.2 can be repeated with D_σ replaced by $\max\left\{\frac{(1+\sigma)^2 C_0 G^2}{\sigma^2}, H^2\right\}$. The maximum with H^2 is what allows the $\max\{0, \cdot\}$ in $(\log)_+$ to be dropped. Lemma 4.1 gives $G = G_\sigma$ and $H = H_\sigma$ from (11), for which the maximum is D_σ. In fact, the proof of Theorem 4.2 used few properties of the outer algorithm in which it was implemented. The main ingredients of the proof are estimates on the drift of the inner subproblem from iteration to iteration. It is an interesting question to see if it is possible to apply this strategy to other well-known algorithms which assume some kind of (possibly inexact) subproblem oracle.

Theorem 4.1 also yields exponential concentration of the total inner work. The proof uses the exponential supermartingale method described by Howard et al. [2020].

Theorem 4.3 (Concentration of total inner work)**.** *We define the upper bound function $\mathcal{T}_\delta$ of Theorem 4.2 as*

$$\begin{aligned} \mathcal{T}_\delta(m) := m \left(\frac{s}{1-\alpha^s} + \frac{\log D_\sigma}{\log(1/\alpha)} \right) \\ + \frac{1}{\log(1/\alpha)} \left(\frac{s \log(1/\alpha)}{1-\alpha^s} + (\log \beta_0)_+ + \log \frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2 \delta^2} \right), \end{aligned}$$

with $(\log 0)_+ := 0$. Under the assumptions of Theorem 4.2, for every $\eta \in (0,1)$,

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta} N_k > \mathcal{T}_\delta(K_\delta) + \frac{\sqrt{2(K_\delta^+ + 1)\log(1/\eta)} + \log(1/\eta)}{\log(1/\alpha)} \right) \le \eta, \tag{15}$$

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta} N_k > \mathcal{T}_\delta(K_\delta) + \frac{K_\delta + 1 + 4\log(1/\eta)}{2\log(1/\alpha)} \right) \le \eta. \tag{16}$$

Proof. We first strengthen Theorem 4.1 to an exponential moment bound. Fix outer iteration k and $0 < \theta < \log(1/\alpha)$. We define

$$l_k = \frac{(\log \beta_k)_+}{s \log(1/\alpha)}, \qquad A_\theta = \frac{\exp(s\theta) - 1}{1 - \alpha^s \exp(s\theta)}.$$

Since $\beta_k \le \exp(s l_k \log(1/\alpha))$ and N_k takes values in $\{0, s, 2s, \dots\}$, summing its conditional tails gives

$$\begin{aligned} \mathbf{E}_k[\exp(\theta N_k)] &= 1 + (\exp(s\theta) - 1) \sum_{j=0}^{\infty} \exp(s\theta j) \mathbf{P}_k(N_k > sj) \\ &\le \exp(s\theta l_k) \left(1 + A_\theta \alpha^{s\lceil l_k \rceil - l_k}\right). \end{aligned}$$

The function $d \mapsto \exp(s\theta d)(1 + A_\theta \alpha^{sd})$ is a sum of two exponentials and is convex on $[0, 1]$. Its values at both endpoints equal $\frac{(1-\alpha^s)\exp(s\theta)}{1-\alpha^s \exp(s\theta)}$. Consequently,

$$\mathbf{E}_k[\exp(\theta(N_k - sl_k))] \le \frac{(1-\alpha^s)\exp(s\theta)}{1-\alpha^s\exp(s\theta)}.$$

Let us define

$$\Gamma(\theta) := \log\left(\frac{(1-\alpha^s)\exp(s\theta)}{1-\alpha^s\exp(s\theta)}\right) - \frac{s\theta}{1-\alpha^s}.$$

Then $\Gamma(0) = \Gamma'(0) = 0$, and

$$\Gamma''(\theta) = \frac{s^2\alpha^s\exp(s\theta)}{(1-\alpha^s\exp(s\theta))^2} = \frac{s^2}{4\sinh^2(s(\log(1/\alpha)-\theta)/2)} \le \frac{1}{(\log(1/\alpha)-\theta)^2}.$$

Integrating twice and expanding the logarithm yields

$$\begin{aligned}\log \mathbf{E}_k\left[\exp\left(\theta\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\right)\right] &\le -\log\left(1 - \frac{\theta}{\log(1/\alpha)}\right) - \frac{\theta}{\log(1/\alpha)} \\ &\le \frac{\theta^2}{2\log(1/\alpha)^2(1-\frac{\theta}{\log(1/\alpha)})} =: \Psi(\theta). \qquad (17)\end{aligned}$$

To account for termination, let $I_k = \mathbf{1}_{\{K_\delta \ge k\}}$ and $Y_k = \left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\mathbf{1}_{\{K_\delta \ge k\}}$, setting $N_k = sl_k = 0$ after termination. The indicator $\mathbf{1}_{\{K_\delta \ge k\}}$ is $\mathcal{F}_k$-measurable. By (17), the process

$$M_j = \exp\left(\sum_{k=0}^{j-1}\left(\theta\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right) - \Psi(\theta)\right)\mathbf{1}_{\{K_\delta \ge k\}}\right), \qquad 0 \le j \le K_\delta^+ + 1,$$

is a nonnegative supermartingale with $M_0 = 1$. Thus, $\mathbf{E}[M_j] \le 1$ for all j. Note that $\sum_{k=0}^{K_\delta^+}\mathbf{1}_{\{K_\delta \ge k\}} = K_\delta + 1 \le K_\delta^+ + 1$, so

$$\begin{aligned}\mathbf{E}\left[\exp\left(\theta\sum_{k=0}^{K_\delta}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\right)\right] &= \mathbf{E}\left[\exp\left(\theta\sum_{k=0}^{K_\delta^+}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\mathbf{1}_{\{K_\delta \ge k\}}\right)\right] \\ &\le \exp((K_\delta^+ + 1)\Psi(\theta)).\end{aligned}$$

Markov's inequality, with

$$\theta = \frac{\log(1/\alpha)\sqrt{2\log(1/\eta)}}{\sqrt{K_\delta^+ + 1} + \sqrt{2\log(1/\eta)}},$$

gives

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right) > \frac{\sqrt{2(K_\delta^+ + 1)\log(1/\eta) + \log(1/\eta)}}{\log(1/\alpha)}\right) \le \exp(-\log(1/\eta)).$$

The pathwise telescoping bound (14) of Theorem 4.2 gives

$$\sum_{k=0}^{K_\delta} sl_k \le \frac{K_\delta \log D_\sigma}{\log(1/\alpha)} + \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2 \log(1/\alpha)}$$
$$\sum_{k=0}^{K_\delta} N_k \le \mathcal{T}_\delta(K_\delta) + \sum_{k=0}^{K_\delta} \left(N_k - sl_k - \frac{s}{1-\alpha^s} \right).$$

Combining the previous inequalities shows that $\sum_{k=0}^{K_\delta} \left(sl_k + \frac{s}{1-\alpha^s} \right) \le \mathcal{T}_\delta(K_\delta)$, proving (15).

Alternatively, applying Markov's inequality directly to $M_{K_\delta^+ + 1}$ gives, for every fixed $0 < \theta < \log(1/\alpha)$,

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta} \left(N_k - sl_k - \frac{s}{1-\alpha^s} \right) > \frac{(K_\delta + 1)\Psi(\theta) + \log(1/\eta)}{\theta} \right) \le \exp(-\log(1/\eta)).$$

Choosing $\theta = \frac{1}{2}\log(1/\alpha)$, for which $\Psi(\theta) = \frac{1}{4}$, proves (16). □

Remark 4.2 (Concentration of the outer stopping time). Replacing $\mathcal{T}_\delta(K_\delta)$ by $\mathcal{T}_\delta(K_\delta^+)$ in (15) gives a deterministic upper bound with confidence $1-\eta$. However, replacing K_δ by $\mathbf{E}[K_\delta]$ in the probability bounds (15) and (16) would require additional control of the outer stopping time. Since $\mathcal{T}_\delta$ is an increasing affine function, a bound on the upper tail of $K_\delta - \mathbf{E}[K_\delta]$ could be combined with Theorem 4.3 by a union bound. The resulting work bound would involve $\mathcal{T}_\delta(\mathbf{E}[K_\delta])$ and an additional deviation term.

Certain parameterized families of problems and inner solvers may admit sharper results with the aforementioned upper tail decay. Establishing concentration about $\mathbf{E}[K_\delta]$ over popular classes of problems and applying it to practical algorithms is an interesting avenue for future work.

Under the assumptions of Theorem 3.2, $\mathbf{E}[K_\delta] = O(\log(1/\delta))$, so the amount of total inner work is $O(\log(1/\delta))$. This highlights the advantage of having a bound in terms of $\mathbf{E}[K_\delta]$, as we did not need to redo any of the outer convergence analysis. An interesting related question is to see if this can be used to formulate an average-case inner work complexity, rather than the worst-case complexity presented here. We leave this direction for future work.

Theorem 4.4 (A deterministic bound)**.** *Suppose we have a deterministic guarantee on the inner procedure of the form:*

$$\|\gamma z^l + y^l - w_k\|^2 \le C_0 \alpha^l \|\gamma z^0 + y^0 - w_k\|^2,$$

where the constants are the same as in Assumption 4.1. Let $D = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}$*, as before, and define*

$$\tilde{\beta}_k = \frac{(1+\sigma)^2 C_0 \|\gamma z^0 + y^0 - w_k\|^2}{\sigma^2 \max\{\|R(w_k)\|^2, \gamma^2\delta^2\}},$$

where (y^0, z^0) *is the initial pair of the kth inner solve. If* $\sigma \in (0,1)$ *and for* $k \ge 1$ *the inner solver is warm-started as in Lemma 4.1, then every inner solve terminates, with*

$$N_k \le s \left\lceil \frac{(\log \tilde{\beta}_k)_+}{s \log(1/\alpha)} \right\rceil \qquad (0 \le k \le K_\delta), \tag{18}$$

and the total inner work satisfies

$$\sum_{k=0}^{K_\delta} N_k \le s \left\lceil \frac{(\log \tilde{\beta}_0)_+}{s \log(1/\alpha)} \right\rceil + K_\delta \left(s + \frac{\log D_\sigma}{\log(1/\alpha)} \right) + \frac{1}{\log(1/\alpha)} \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}. \tag{19}$$

Proof. Fix an outer iteration k and a graph pair (y, z) with residual $e = \gamma z + y - w_k$. The proof of Theorem 4.1 shows that if both tests of Algorithm 1 fail at (y, z), then $\|e\| > \frac{\sigma}{1+\sigma} \max\{\|R(w_k)\|, \gamma\delta\}$. That argument used only (5) and the triangle inequality, so no randomness is involved. Hence one of the tests passes as soon as $\|e\| \le \frac{\sigma}{1+\sigma} \max\{\|R(w_k)\|, \gamma\delta\}$.

At the check after sl inner iterations, the deterministic guarantee and an argument similar to Theorem 4.1 give

$$\begin{aligned}\|\gamma z^{sl} + y^{sl} - w_k\|^2 &\le C_0 \alpha^{sl} \|\gamma z^0 + y^0 - w_k\|^2 \\ &= \tilde{\beta}_k \alpha^{sl} \left(\frac{\sigma}{1+\sigma}\right)^2 \max\{\|R(w_k)\|, \gamma\delta\}^2.\end{aligned}$$

By the first paragraph, the inner loop exits by the first check l with $\tilde{\beta}_k \alpha^{sl} \le 1$, and $l \ge \lceil \frac{(\log \tilde{\beta}_k)_+}{s \log(1/\alpha)} \rceil$ satisfies this. This proves (18), and in particular $N_k \le s + \frac{(\log \tilde{\beta}_k)_+}{\log(1/\alpha)}$.

For $1 \le k \le K_\delta$, the inequalities of Lemma 4.1 and Corollary 4.1.1 also hold deterministically, with $\tilde{\beta}_k$ in place of β_k. The argument in the proof of Theorem 4.2 therefore gives the analog of (14) with $K = K_\delta$:

$$\begin{aligned}\sum_{k=1}^{K_\delta} N_k &\le K_\delta s + \frac{1}{\log(1/\alpha)} \sum_{k=1}^{K_\delta} (\log \tilde{\beta}_k)_+ \\ &\le K_\delta \left(s + \frac{\log D_\sigma}{\log(1/\alpha)}\right) + \frac{1}{\log(1/\alpha)} \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}.\end{aligned}$$

Adding the bound (18) for $k = 0$ gives the desired inequality in (19). □

Remark 4.3. Compared with Theorem 4.2, the per-iteration constant $\frac{s}{1-\alpha^s}$ improves to s, because there is no random tail to sum. Remark B.1 in Appendix A shows that the per-step bound (18) is attained, that the expectation bound of Theorem 4.1 can be approached arbitrarily closely, and that, for $\sigma \le \sigma_1$, the constant D_σ in the total bounds is sharp up to the factor $(1+\sigma)^2$.

The same inner-work analysis also allows relaxation of the outer update. The graph condition, relative error test, and terminal test remain unchanged.

Proposition 4.1 (Relaxed inexact PPM)**.** *Suppose Assumptions 2.2 and 4.1 hold, and fix*

$$0 < \theta < 2, \qquad 0 < \sigma < \min\left\{1, \frac{2-\theta}{\theta}\right\}.$$

Consider Algorithm 1 with its accepted update replaced by

$$w_{k+1} = w_k - \theta\gamma z_k.$$

Define

$$c_{\sigma,\theta} := \theta\left(\frac{2}{1+\sigma} - \theta\right) > 0.$$

Every accepted update satisfies, for any $w^\star \in \mathbf{zer}\, B$,

$$\|w_{k+1} - w^\star\|^2 \le \|w_k - w^\star\|^2 - c_{\sigma,\theta}\gamma^2 \|z_k\|^2. \tag{20}$$

Consequently, for $\delta > 0$, the number of accepted updates before the terminal test passes obeys almost surely

$$K_\delta \le \left\lfloor \frac{\mathbf{dist}(w_0, \mathbf{zer}\, B)^2}{c_{\sigma,\theta}\gamma^2\delta^2} \right\rfloor. \tag{21}$$

The terminal certificate of Proposition 2.1 is unchanged.

If the inner solver reuses the previously accepted graph pair, set

$$G_{\sigma,\theta} := \frac{\sigma+\theta}{1-\sigma}, \qquad H_{\sigma,\theta} := 1 + \frac{\theta}{1-\sigma}, \qquad D_{\sigma,\theta} := \frac{(1+\sigma)^2 C_0 G_{\sigma,\theta}^2}{\sigma^2}.$$

Then the expected total-work bound and the high-probability bounds of Theorems 4.2 and 4.3 hold with D_σ replaced by $D_{\sigma,\theta}$ and K_δ denoting the relaxed outer stopping time (21). With a deterministic inner contraction, the same replacement gives Theorem 4.4. In particular, the expected total inner work remains $O(\delta^{-2})$ for fixed admissible parameters and initialization. This bound holds deterministically when the inner contraction is deterministic.

The proof is given in Appendix C. The restriction $\theta < \frac{2}{1+\sigma}$ is necessary for a uniform convergence guarantee with this relative test. For example, take $B = N_{\{0\}}$, $\gamma = 1$, $y = 0$, and $z = (1+\sigma)w$. The relative test holds with equality, while $w^+ = (1-\theta(1+\sigma))w$. For nonzero w, equality $\theta(1+\sigma) = 2$ produces a two-cycle, and larger values give divergence. With exact solves, $\sigma = 0$ and the decrease coefficient reduces to the well-known $\theta(2-\theta)$.

One might wonder if Assumption 4.1 is strong enough to ensure that every inner loop terminates when the terminal test of Algorithm 1 is removed, i.e., when only the relative error test is used. We show with a simple example that this is not the case. This phenomenon has already been noted in Eckstein and Yao [2018].

Example 4.1 (An Infinite Inner Loop). Suppose $\mathcal{H} = \mathcal{D} = \mathbb{R}$, $C = 1$, $\gamma = 1$, and $A = I$. This is equivalent to minimizing $f(x) = \frac{1}{2}x^2$. A valid graph pair has $z = u = y$ and error $e = z + y - w = 2u - w$. At the exact outer fixed point $w = 0$, take $u^l = z^l = \alpha^{l/2}$ for $0 < \alpha < 1$, so $u^0 = 1$ (superscripts denote inner iterations, as throughout). Then $|e^l|^2 = 4\alpha^l = \alpha^l|e^0|^2$, but

$$|e^l| = 2|u^l| > \sigma|u^l - w| \qquad (0 < \sigma < 1)$$

at every finite index. Consequently a purely relative inner loop never terminates, even under deterministic geometric contraction. The terminal condition $|z^l| = \alpha^{l/2} \le \delta$ does terminate for every $\delta > 0$. This shows why merely declaring the outer point to be a fixed point, or invoking convergence of e^l to zero, does not justify finite inner termination.

4.2 Selection of Parameters

We now use Theorem 4.2 to choose the stride s, given the costs of an inner iteration and of a check. The choice of σ depends on the inner solver and is examined for specific methods in Propositions 5.1 and 5.2.

Lemma 4.2. *Suppose the cost of testing is c_{test}, the cost of each inner iteration is c_{in}, the cost of each outer iteration is c_{out}, and the cost of initialization is c_{init}. Given an upper bound $K_{\text{up}} \ge \mathbf{E}[K_\delta]$, under the assumptions of Theorem 4.2, we can bound the total cost M as*

$$\mathbf{E}[M] \le c_{\text{init}} + c_{\text{out}}(K_{\text{up}} + 1) + c_{\text{in}}N_{\text{tot}} + c_{\text{test}}(K_{\text{up}} + 1 + s^{-1}N_{\text{tot}}),$$

where

$$N_{\text{tot}} := (K_{\text{up}} + 1)\left(s + \frac{1}{\log(1/\alpha)}\right) + N',$$
$$N' := \frac{1}{\log(1/\alpha)}\left(K_{\text{up}} \log D_\sigma + (\log \beta_0)_+ + \log \frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2}\right),$$

and D, β_0 are as in Theorem 4.2. The choice of stride $s \in \mathbb{N}$ that minimizes this upper bound is one of $\lfloor s^\star \rfloor$ or $\lceil s^\star \rceil$, where

$$s^\star := \sqrt{\frac{c_{\text{test}}\left(N' + \frac{K_{\text{up}}+1}{\log(1/\alpha)}\right)}{c_{\text{in}}(K_{\text{up}}+1)}}.$$

Proof. Since $\frac{s}{1-\alpha^s} \le s + \frac{1}{\log(1/\alpha)}$ and $\mathbf{E}[K_\delta] \le K_{\text{up}}$, Theorem 4.2 gives $\mathbf{E}[\sum_{k=0}^{K_\delta} N_k] \le N_{\text{tot}}$. Outer iteration k performs one test before any inner iteration and one after every s inner iterations, i.e., $1 + N_k/s$ tests. There are $K_\delta + 1$ outer iterations, so

$$\begin{aligned}
\mathbf{E}[M] &= \mathbf{E}\left[c_{\text{init}} + \sum_{k=0}^{K_\delta}\left[c_{\text{out}} + c_{\text{in}}N_k + c_{\text{test}}(1 + s^{-1}N_k)\right]\right] \\
&\le c_{\text{init}} + c_{\text{out}}(K_{\text{up}}+1) + c_{\text{in}}N_{\text{tot}} + c_{\text{test}}(K_{\text{up}} + 1 + s^{-1}N_{\text{tot}}) \qquad (22) \\
&= M' + c_{\text{in}}(K_{\text{up}}+1)\, s + c_{\text{test}}\left(N' + \frac{K_{\text{up}}+1}{\log(1/\alpha)}\right) s^{-1},
\end{aligned}$$

where M' suppresses the terms not involving the stride s. Writing $M_1 := c_{\text{in}}(K_{\text{up}}+1)$ and $M_2 := c_{\text{test}}\left(N' + \frac{K_{\text{up}}+1}{\log(1/\alpha)}\right)$, the right-hand side is $M' + M_1 s + M_2 s^{-1}$ with $M', M_1, M_2 \ge 0$. As a function of s, it is convex on $(0,\infty)$ and minimized at $s^\star = \sqrt{M_2/M_1}$. Since it is decreasing on $(0, s^\star]$ and increasing on $[s^\star, \infty)$, its minimizer over the positive integers is one of $\lfloor s^\star \rfloor$ or $\lceil s^\star \rceil$. □

Remark 4.4. The only relaxation in this bound beyond Theorem 4.2 is the use of the inequality $\frac{s}{1-\alpha^s} \le s + \frac{1}{\log(1/\alpha)}$, which costs at most $\frac{s}{2}$ inner iterations per outer iteration. It is a heuristic that one could reasonably estimate before launching their algorithm of choice. For an exact answer, one would replace $s + \frac{1}{\log(1/\alpha)}$ in N_{tot} in (22) with $\frac{s}{1-\alpha^s}$ and attempt to numerically find the minimum.

5 Application to First-Order Methods

In general, the relationship between parameters is complex and depends on a variety of factors. Moreover, each choice of algorithm requires a separate analysis of how to optimally choose the parameters σ, s, and γ. In this section, we consider a few common methods and propose heuristics for selecting parameters. Note that these heuristics are oriented toward reduction of the fixed point residual, which may not be the metric of interest in all cases. Nevertheless, we hope this provides a more principled heuristic for implementations of these algorithms.

5.1 Proximal Point Method

As a prescriptive example, let us consider the case of inexact proximal point on a smooth convex function. The optimal choice of fixed stepsize is $\gamma_{\text{GD}} = \frac{2}{(L+\frac{1}{\gamma})+\frac{1}{\gamma}} = \frac{2\gamma}{\gamma L+2}$, where γ_{GD} is the inner stepsize of the gradient method applied to the proximal subproblem $\min_x \left\{f(x) + \frac{1}{2\gamma}\|x - w\|^2\right\}$, which is $(L + \frac{1}{\gamma})$-smooth and $(\frac{1}{\gamma})$-strongly convex.

Proposition 5.1. *Consider the case where $C = I$ and $A = \nabla f$ where f is convex and L-smooth. At outer iteration k, suppose the inner procedure is gradient descent with stepsize $\gamma_{\mathrm{GD}} = \frac{2\gamma}{\gamma L+2}$ on the proximal subproblem $\operatorname{argmin}_x \varphi_k(x)$, where $\varphi_k(x) := f(x) + \frac{1}{2\gamma}\|x - w_k\|^2$, warm-started at the previously accepted u, and uses the graph pair $(u^l, z^l) = (x^l, \nabla f(x^l))$. Then, the following hold:*

1. *The deterministic guarantee of Theorem 4.4, and hence Assumption 4.1, holds with*

$$\alpha = \left(\frac{\gamma L}{\gamma L + 2}\right)^2, \qquad C_0 = (1 + \gamma L)^2.$$

2. *Let $\chi := \frac{s\log(1/\alpha)}{1-\alpha^s}$ for the bound of Theorem 4.2, or $\chi := s\log(1/\alpha)$ for the bound of Theorem 4.4. Substitute $\mathbf{E}[K_\delta] \le K_\delta^+ \le \frac{1+\sigma}{1-\sigma} \cdot \frac{\mathbf{dist}(w_0, \mathbf{zer}\, B)^2}{\gamma^2\delta^2}$ into either bound. The coefficient of δ^{-2} is then $\frac{\mathbf{dist}(w_0,\mathbf{zer}\, B)^2}{\gamma^2\log(1/\alpha)} g(\sigma)$, where*

$$g(\sigma) := \frac{1+\sigma}{1-\sigma}\left(\chi + \log D_\sigma(\sigma)\right), \qquad D(\sigma) = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}.$$

Over $\sigma \in (0,1)$, g has a unique minimizer $\sigma^\star$, which lies in $(0, \sigma_1)$ and is the unique solution in $(0, \sigma_1)$ of

$$\frac{1}{\sigma} = 1 + \psi(\sigma) + \chi + \log\frac{(1+\sigma)^2(1+2\sigma)^2 C_0}{\sigma^2}, \qquad \psi(\sigma) := \frac{2(1-\sigma^2)}{1+2\sigma}. \tag{23}$$

Moreover,

$$\sigma^\star < \frac{1}{3/2 + s\log(1/\alpha) + 2\log(1+\gamma L)}.$$

The other σ-dependent term in the bounds, $\frac{(\log\beta_0)_+}{\log(1/\alpha)}$, does not grow as $\delta \to 0$. Consequently, $\sigma^\star$ minimizes the upper bound to leading order in δ^{-2}.

Proof. Here $B = \nabla f$, $y = u$, and $J = (I + \gamma\nabla f)^{-1}$ maps w_k to the minimizer $x_k^\star$ of φ_k. The pair $(x^l, \nabla f(x^l))$ lies in $\mathbf{gra}\, B$, and its residual is $e^l = \gamma\nabla f(x^l) + x^l - w_k = \gamma\nabla\varphi_k(x^l)$. The function φ_k is μ-strongly convex and L_φ-smooth with $\mu = \frac{1}{\gamma}$ and $L_\varphi = L + \frac{1}{\gamma}$, so its condition number is $\frac{L_\varphi}{\mu} = 1 + \gamma L$. With stepsize $\frac{2}{L_\varphi+\mu} = \gamma_{\mathrm{GD}}$, gradient descent satisfies $\|x^l - x_k^\star\| \le \alpha^{l/2}\|x^0 - x_k^\star\|$ with $\alpha = \left(\frac{L_\varphi-\mu}{L_\varphi+\mu}\right)^2 = \left(\frac{\gamma L}{\gamma L+2}\right)^2$ [Nesterov, 2013, Theorem 2.1.15]. Since $\mu\|x - x_k^\star\| \le \|\nabla\varphi_k(x)\| \le L_\varphi\|x - x_k^\star\|$,

$$\begin{aligned}
\|e^l\|^2 &= \gamma^2\|\nabla\varphi_k(x^l)\|^2 \\
&\le \gamma^2 L_\varphi^2 \alpha^l \|x^0 - x_k^\star\|^2 \\
&\le \left(\frac{L_\varphi}{\mu}\right)^2 \alpha^l \gamma^2 \|\nabla\varphi_k(x^0)\|^2 \\
&= (1+\gamma L)^2 \alpha^l \|e^0\|^2.
\end{aligned}$$

This is a deterministic bound that holds for every inner iteration l, independently of outer iteration k, so we are in the setting of Theorem 4.4.

Define $S = \frac{1+\sigma}{1-\sigma}$ and $\Delta = \frac{\mathbf{dist}(w_0, \mathbf{zer}\, B)^2}{\gamma^2\delta^2}$. The bound of Theorem 4.2 is at most

$$\frac{S\Delta}{\log(1/\alpha)}(\chi + \log D_\sigma(\sigma)) + \frac{s}{1-\alpha^s} + \frac{1}{\log(1/\alpha)}\left((\log\beta_0)_+ + \log\frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}\right).$$

The bound of Theorem 4.4 has the same δ^{-2} term with $\chi = s\log(1/\alpha)$, and its remaining terms also do not grow with δ^{-1} faster than $\log(1/\delta)$. This identifies the δ^{-2} coefficient as claimed. Here α, C_0, and hence χ do not depend on σ.

Let $\tilde{D}(\sigma) := \frac{(1+\sigma)^2(1+2\sigma)^2 C_0}{\sigma^2}$ and $\tilde{g}(\sigma) := S(\sigma)\left(\chi + \log\tilde{D}(\sigma)\right)$. By Lemma 4.1, $G_\sigma \geq 1 + 2\sigma$ with equality for $\sigma \leq \sigma_1$, so $g \geq \tilde{g}$ on $(0,1)$, with equality on $(0, \sigma_1]$. We first minimize $\tilde{g}$. Since $\log\tilde{D} = \log C_0 + 2\log(1+\sigma) + 2\log(1+2\sigma) - 2\log\sigma$,

$$\frac{1-\sigma^2}{2}\frac{d}{d\sigma}\log\tilde{D} = 1 + \psi(\sigma) - \frac{1}{\sigma},$$

with $\psi(\sigma) = \frac{2(1-\sigma^2)}{1+2\sigma}$ as in the optimality condition (23). Using $\frac{d}{d\sigma}\log S = \frac{2}{1-\sigma^2}$, we get

$$\begin{aligned}\tilde{g}'(\sigma) &= \frac{2S}{1-\sigma^2}\left(\chi + \log\tilde{D}(\sigma) + 1 + \psi(\sigma) - \frac{1}{\sigma}\right)\\ &= -\frac{2S}{1-\sigma^2}\,h(\sigma),\\ h(\sigma) &:= \frac{1}{\sigma} - 1 - \psi(\sigma) - \chi - \log\tilde{D}(\sigma).\end{aligned}$$

Now $h(\sigma) \to +\infty$ as $\sigma \downarrow 0$, since $1/\sigma$ dominates $\log\tilde{D}(\sigma) = 2\log(1/\sigma) + O(1)$, and $h(1) = -\chi - \log(36C_0) < 0$. Moreover, h is strictly decreasing: since $\psi'(\sigma) = -\frac{4(\sigma^2+\sigma+1)}{(1+2\sigma)^2}$,

$$h'(\sigma) = -\frac{1}{\sigma^2} + \frac{2}{\sigma} - \frac{2}{1+\sigma} - \frac{4}{1+2\sigma} + \frac{4(\sigma^2+\sigma+1)}{(1+2\sigma)^2} = -\frac{(1-\sigma)^2}{\sigma^2} - \frac{1-\sigma}{1+\sigma} - \frac{4\sigma(1-\sigma)}{(1+2\sigma)^2} < 0.$$

Hence h has a unique root $\sigma^\star \in (0,1)$, which solves (23), and $\tilde{g}$ is strictly decreasing on $(0, \sigma^\star)$ and strictly increasing on $(\sigma^\star, 1)$. Furthermore $\sigma^\star < \sigma_1$, because $h(\sigma_1) < 0$: indeed $\psi(\sigma_1) = \frac{1}{2}$ and $\frac{1}{\sigma_1} - 1 - \psi(\sigma_1) = \frac{1+\sqrt{13}}{3} - \frac{3}{2} < 0.04$, while $\log\tilde{D}(\sigma_1) = \log C_0 + 2\log\frac{17+5\sqrt{13}}{6} > \log C_0 + 2\log 5 > 3$. It follows that $\sigma^\star$ is also the unique minimizer of g: for $\sigma \leq \sigma_1$, $g(\sigma) = \tilde{g}(\sigma) \geq \tilde{g}(\sigma^\star) = g(\sigma^\star)$ with equality only at $\sigma = \sigma^\star$, and for $\sigma > \sigma_1$, $g(\sigma) \geq \tilde{g}(\sigma) > \tilde{g}(\sigma_1) \geq \tilde{g}(\sigma^\star) = g(\sigma^\star)$. Finally, $\log\tilde{D}(\sigma) > \log C_0$ on $(0,1)$, and ψ is decreasing, so $\psi(\sigma^\star) \geq \psi(\sigma_1) = \frac{1}{2}$. Substituting into (23) gives $\frac{1}{\sigma^\star} > \frac{3}{2} + \chi + \log C_0$. □

Proposition 5.2. *Consider the setting of Proposition 5.1, with $L > 0$. Define $\kappa := 1 + \gamma L$, $L_\varphi := L + \frac{1}{\gamma}$, and $\rho := \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$. At each outer iteration, apply Nesterov's accelerated gradient method to φ_k, with x^0 warm-started at the previously accepted u and $x^{-1} = x^0$:*

$$\begin{aligned}v^l &= x^l + \rho(x^l - x^{l-1}),\\ x^{l+1} &= v^l - \frac{1}{L_\varphi}\nabla\varphi_k(v^l), \qquad l = 0, 1, 2, \ldots.\end{aligned}$$

Note that momentum is reset at the beginning of each inner solve. Use the graph pair $(u^l, z^l) = (x^l, \nabla f(x^l))$ at each stopping test. Then the following hold:

1. *The deterministic guarantee of Theorem 4.4, and hence Assumption 4.1, holds with*

$$\alpha = 1 - \frac{1}{\sqrt{\kappa}}, \qquad C_0 = 2\kappa. \tag{24}$$

2. *Let $\chi := \frac{s\log(1/\alpha)}{1-\alpha^s}$ for the expectation bound, or $\chi := s\log(1/\alpha)$ for the deterministic bound. After substituting the outer iteration bound as in Proposition 5.1, the coefficient of δ^{-2} is*

$$\frac{\mathbf{dist}(w_0, \mathbf{zer}\, B)^2}{\gamma^2 \log(1/\alpha)} g(\sigma), \qquad g(\sigma) := \frac{1+\sigma}{1-\sigma}\left(\chi + \log\frac{2\kappa(1+\sigma)^2 G_\sigma^2}{\sigma^2}\right).$$

The function g has a unique minimizer $\sigma^\star \in (0, \sigma_1)$, where $\sigma_1 = \frac{\sqrt{13}-1}{4}$. It is the unique solution of (23) in this interval with $C_0 = 2\kappa$ and the above choice of χ. Moreover,

$$\sigma^\star < \frac{1}{3/2 + \chi + \log(2\kappa)} \le \frac{1}{3/2 + s\log(1/\alpha) + \log(2\kappa)}.$$

For fixed initialization and algorithm parameters, the remaining terms grow at most as $\log(1/\delta)$ as $\delta \downarrow 0$. Consequently, $\sigma^\star$ minimizes the upper bound to leading order in δ^{-2}.

The proof is given in Appendix D, as it is extremely similar to the proof of Proposition 5.1.

Remark 5.1. Note that the inner iterations of Proposition 5.1 are of the form $x_k^{l+1} = x_k^l - \gamma_{\mathrm{GD}}(\nabla f(x_k^l) + \frac{1}{\gamma}(x_k^l - w_k))$. That is, all future iterates are in the span of past iterates and gradients. Any such method needs $\sum_{k=0}^{K_\delta} N_k \gtrsim \delta^{-1/2}$. This follows[2] from analyzing "the worst function in the world" [Nesterov, 2013, Section 2.1.2]. One can see a similar property for Proposition 5.2. We leave a more detailed analysis of lower bounds future work.

5.2 Douglas–Rachford

Alves et al. [2026] show that the inexact Douglas–Rachford splitting of Eckstein and Yao [2018] fits into their framework. We reproduce the formulation here. Suppose we wish to solve

$$\text{find } x \text{ such that } 0 \in A_1 x + A_2 x, \tag{25}$$

where A_1 and A_2 are maximal monotone operators. We introduce the operators

$$A = \begin{bmatrix} \tau A_1 & I & -I \\ -I & \tau A_2 & I \\ I & -I & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} I \\ -I \\ I \end{bmatrix}, \qquad M = CC^*, \tag{26}$$

and set $\gamma = 1$. One can check that $(CC^* + \gamma A)^{-1} CC^*$ is defined everywhere. Note that if $0 \in A(x_1, x_2, v)$, then $x_1 = x_2 =: x$ and $0 \in A_1 x + A_2 x$. Substituting this into Algorithm 1 yields Algorithm 2. We also remark that this is not the same formulation as that of Eckstein and Bertsekas [1992], who show that Douglas–Rachford splitting is an instance of PPM for a certain splitting operator. However, the splitting operator is difficult to define and understand. This formulation shows that Douglas–Rachford splitting can be viewed as the projection of PPM done in a product space. In Appendix E we show that Algorithm 2 is an instance of Algorithm 1 with A and C as in (26). Alternatively, Proposition 5.4, whose proof is in Appendix G, with $A_3 = 0$ proves that Algorithm 2 is an instance of Algorithm 1.

At termination, the returned quantities satisfy $u_k \in A_1 x_k$, $v_k \in A_2 y_k$, $\|x_k - y_k\| \le \delta$, and $\|u_k + v_k\| = \frac{1}{\tau}\|x_k - y_k\| \le \frac{\delta}{\tau}$. That is, (x_k, y_k) solves (25) up to δ in the primal variables and $\frac{\delta}{\tau}$ in the operator values.

[2] Namely, using Theorem 3.1 to conclude that gradients and iterates are bounded by initialization constants.

Algorithm 2 The method introduced in Eckstein and Yao [2018] (adapted from [Alves et al., 2026, Algorithm 2]) as an instance of Algorithm 1.

1: **Initialize:** $w_0 \in \mathcal{H}$, stepsize $\tau > 0$, inexactness $\sigma \in [0, 1)$, stride $s \in \mathbb{N}$, and stopping tolerance δ
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Produce a pair $(x_k, u_k) \in \mathcal{H} \times \mathcal{H}$ with

$$u_k \in A_1 x_k, \qquad \tau u_k + x_k \approx w_k.$$

4: Compute $y_k = J_{\tau A_2}(x_k - \tau u_k)$ and $v_k = \frac{1}{\tau}(x_k - \tau u_k - y_k)$, so that

$$v_k \in A_2 y_k, \qquad \tau v_k + y_k = x_k - \tau u_k.$$

5: **if** $\|x_k - y_k\| \leq \delta$ **then**
6: Stop and return (x_k, y_k, u_k, v_k).
7: **else if** $\|x_k + \tau u_k - w_k\| \leq \sigma \|y_k + \tau u_k - w_k\|$ **then**
8: Set $w_{k+1} = w_k - \tau(u_k + v_k)$ (equivalently, $w_{k+1} = w_k + y_k - x_k$).
9: **else**
10: Improve the pair (x_k, u_k) with $u_k \in A_1 x_k$ to reduce $\|x_k + \tau u_k - w_k\|$ via the iterative procedure s times and go to line 4.
11: **end if**
12: **end for**

5.3 ADMM

We consider the linearly constrained composite optimization problem

$$\begin{array}{ll} \text{minimize} & f(x) + g(z) \\ \text{subject to} & Ex = z, \end{array} \tag{27}$$

where $x \in \mathbb{R}^n$ and $z \in \mathbb{R}^m$ are the optimization variables, $E \in \mathbb{R}^{m \times n}$, and f and g are closed, proper, and convex functions. We assume that the problem (27) has a solution. That is, there exists a tuple (x, z, a, b) satisfying the KKT conditions:

$$Ex = z, \qquad a + E^\top b = 0, \qquad a \in \partial f(x), \quad b \in \partial g(z).$$

We obtain a fully inexact version and a version with a single inexact solve of the proximal ADMM method of Sun et al. [2025]. For $c > 0$ and $\rho > 0$, define

$$A_E = \begin{bmatrix} \frac{1}{c}\partial f & E^\top & -E^\top \\ -E & \frac{1}{c}\partial g & I \\ E & -I & 0 \end{bmatrix}, \qquad C_{E,\rho} = \begin{bmatrix} E^\top & \sqrt{\rho} I \\ -I & 0 \\ I & 0 \end{bmatrix}, \qquad \gamma = 1. \tag{28}$$

For the case $\rho = 0$, we must additionally assume that E has full column rank and take $C_{E,\rho}$ as

$$C_{E,0} = \begin{bmatrix} E^\top \\ -I \\ I \end{bmatrix}. \tag{29}$$

The operator A_E is unchanged. Note that in either case,

$$C_{E,\rho}C_{E,\rho}^\top = \begin{bmatrix} E^\top E + \rho I & -E^\top & E^\top \\ -E & I & -I \\ E & -I & I \end{bmatrix}.$$

The ρI term induces a proximal term in the first subproblem. This allows both subproblems to be solved inexactly when $\rho > 0$. We remark that one may replace the $\sqrt{\rho}I$ term in $C_{E,\rho}$ by any strictly positive definite matrix.

We maintain iterates (z_k, p_k, t_k, w_k), with $w_k = z_k - \frac{1}{c}p_k$. We also define $W_k = (w_k, \sqrt{\rho}t_k)$. The subproblem objectives are

$$\varphi_k(x) := f(x) + \frac{c}{2}\|Ex - w_k\|^2 + \frac{c\rho}{2}\|x - t_k\|^2, \tag{30a}$$

$$\psi_k(z; x) := g(z) + \frac{c}{2}\|z - (2Ex - w_k)\|^2. \tag{30b}$$

When $\rho = 0$, the proximal term in φ_k vanishes. The inner procedure finds $a_k \in \partial f(x_k)$ and $\ell_k \in \mathbb{R}^m$ with $E^\top \ell_k = a_k$. It also solves the second subproblem exactly:

$$z_{k+1} = \operatorname{prox}_{g/c}(Ex_k - \frac{1}{c}\ell_k), \qquad b_k = c(Ex_k - z_{k+1}) - \ell_k. \tag{31}$$

Full column rank ensures that such an ℓ_k exists.

The inner procedure returns graph points $a_k \in \partial f(x_k)$ and $b_k \in \partial g(z_{k+1})$. These inclusions remain exact even when the subproblem optimality equations are inexact. For example, when g is the indicator of a closed convex set K, the second graph point must satisfy $z_{k+1} \in K$ and $b_k \in N_K(z_{k+1})$, where N_K is the normal-cone operator. Define

$$d_k = Ex_k - z_{k+1}, \tag{32a}$$

$$q_k = \frac{a_k + E^\top b_k}{c} - E^\top d_k, \tag{32b}$$

$$r_{f,k} = \frac{a_k}{c} + E^\top(Ex_k - w_k) + \rho(x_k - t_k), \tag{32c}$$

$$r_{g,k} = z_{k+1} + \frac{b_k}{c} - 2Ex_k + w_k. \tag{32d}$$

The last two quantities are rescaled first-order residuals of the subproblems (30):

$$r_{f,k} \in \frac{1}{c}\partial\varphi_k(x_k), \qquad r_{g,k} \in \frac{1}{c}\partial_z\psi_k(z_{k+1}; x_k). \tag{33}$$

For $\rho > 0$, the terminal and relative error tests use the nonnegative quantities

$$Q_k^2 = \|d_k\|^2 + \frac{1}{\rho}\|q_k\|^2, \tag{34a}$$

$$E_k^2 = \|r_{g,k}\|^2 + \frac{1}{\rho}\|r_{f,k} + E^\top r_{g,k}\|^2, \tag{34b}$$

$$P_k^2 = \|d_k + r_{g,k}\|^2 + \rho\|x_k - t_k\|^2. \tag{34c}$$

When $\rho = 0$, the exact solve (31) enforces $q_k = 0$. In this case, define

$$Q_k = \|Ex_k - z_{k+1}\|, \tag{35a}$$

$$E_k = \|Ex_k + \frac{1}{c}\ell_k - w_k\|, \tag{35b}$$

$$P_k = \|z_{k+1} + \frac{1}{c}\ell_k - w_k\|. \tag{35c}$$

The quantity E_k measures the inexactness of the first solve together with its lift. Note that Algorithm 3 does not require either subproblem to be solved exactly when $\rho > 0$. Substituting these quantities into Algorithm 1 gives Algorithm 3.

Algorithm 3 Doubly inexact proximal ADMM as an instance of Algorithm 1.

1: **Initialize:** z_0, p_0, t_0, penalty $c > 0$, proximal parameter $\rho \geq 0$, inexactness $\sigma \in [0, 1)$, stride $s \in \mathbb{N}$, and tolerance $\delta > 0$.
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Set $w_k = z_k - \frac{1}{c} p_k$ and initialize graph points $a_k \in \partial f(x_k)$ and $b_k \in \partial g(z_{k+1})$ if $\rho > 0$, or (x_k, a_k, ℓ_k) with $E^\top \ell_k = a_k \in \partial f(x_k)$ if $\rho = 0$.
4: For $k \geq 1$, reuse $(x_{k-1}, z_k, a_{k-1}, b_{k-1})$ if $\rho > 0$, or $(x_{k-1}, a_{k-1}, \ell_{k-1})$ if $\rho = 0$, before the first check to warm start the inner procedure.
5: If $\rho = 0$, compute (z_{k+1}, b_k) by the exact solve (31).
6: Compute $d_k, q_k, r_{f,k}, r_{g,k}$ and Q_k, E_k, P_k from (32) and (34) when $\rho > 0$, or using (35) instead when $\rho = 0$.
7: **if** $Q_k \leq \delta$ **then**
8: Stop and return (x_k, z_{k+1}, a_k, b_k).
9: **else if** $E_k \leq \sigma P_k$ **then**
10: Keep z_{k+1} and set $p_{k+1} = p_k + c(Ex_k - z_k)$ and, if $\rho > 0$, $t_{k+1} = t_k - \frac{1}{\rho} q_k$.
11: **else**
12: Improve iterates through s steps of the inner procedure, preserving subgradient inclusions. In addition, when $\rho = 0$, also preserve $E^\top \ell_k = a_k$. Return to line 5.
13: **end if**
14: **end for**

For $\rho > 0$, expanding the subproblem subgradients of (33) shows that the auxiliary t_k update can also be written as

$$t_{k+1} = x_k - \frac{r_{f,k} + E^\top r_{g,k}}{\rho}.$$

When $E_k = 0$ (which is guaranteed when $\sigma = 0$), the method reduces to proximal ADMM of Sun et al. [2025]. Setting $\rho = 0$ and $E = I$ yields the inexact ADMM algorithm of Eckstein and Yao [2018].

Proposition 5.3. *Suppose f and g are closed, convex, and proper functions, and that the composite optimization problem (27) has a solution. Let $c > 0$ and $\rho \geq 0$ be the penalty and proximal parameters, respectively. If $\rho = 0$, additionally assume that E has full column rank. Then the following hold:*

1. *The operators in (28) (or (29) when $\rho = 0$) satisfy Assumption 2.1 with $\gamma = 1$. Algorithm 3 is Algorithm 1 applied to A_E and $C_{E,\rho}$. Hence, the outer convergence results of Section 3 apply here. The results of Section 4 apply whenever the inner solver contracts E_k^2 linearly in expectation.*

2. *At termination, $a_k \in \partial f(x_k)$ and $b_k \in \partial g(z_{k+1})$, with*

$$\|Ex_k - z_{k+1}\| \leq \delta, \qquad \|a_k + E^\top b_k\| \leq c\delta(\|E\| + \sqrt{\rho}).$$

That is, the pair (x_k, z_{k+1}) satisfies the KKT conditions of (27) up to δ in feasibility and $c\delta(\|E\| + \sqrt{\rho})$ in the first-order condition.

The proof of Proposition 5.3 is in Appendix F.

5.3.1 SDP Application

Consider the semidefinite program (SDP)

$$\begin{array}{ll} \text{minimize} & \langle C, X\rangle \\ \text{subject to} & E(X) = b, \\ & X \succeq 0, \end{array}$$

where $X \in \mathbb{S}^n$ is the optimization variable, $E : \mathbb{S}^n \to \mathbb{R}^m$ is a linear map, $C \in \mathbb{S}^n$, and $b \in \mathbb{R}^m$. We can rewrite this as

$$\begin{array}{ll} \text{minimize} & \langle C, X\rangle + \mathcal{I}_{\{y|y=b\}}(y) + \mathcal{I}_{\{Z|Z\succeq 0\}}(Z) \\ \text{subject to} & E(X) = y, \\ & X = Z, \end{array}$$

where we introduced new optimization variables $Z \in \mathbb{S}^n$ and $y \in \mathbb{R}^m$. Letting $f(X) = \langle C, X\rangle$ and $g(y, Z) = \mathcal{I}_{\{y|y=b\}}(y) + \mathcal{I}_{\{Z|Z\succeq 0\}}(Z)$, we can fit this into the doubly inexact proximal ADMM Algorithm 3. The main difficulty is the projection onto the positive semidefinite cone and solving the f-subproblem. As the f-subproblem is an unconstrained strongly convex quadratic, it is equivalent to solving a positive definite linear system. Note that Algorithm 3 requires producing a subgradient of g at the candidate point (y_{k+1}, Z_{k+1}). For projection onto the singleton set $\{b\}$, its normal cone, equivalently the subgradient of $\mathcal{I}_{\{y|y=b\}}$, is $\mathbb{R}^m$. It is known that

$$N_{\{Z|Z\succeq 0\}}(Z) = \begin{cases} \{0\}, & Z \succ 0 \\ \{A \mid A \preceq 0, \mathbf{Tr}(AZ) = 0\}, & Z \succeq 0, Z \not\succ 0 \\ \varnothing, & Z \not\succeq 0. \end{cases}$$

Therefore, a method like (block) Lanczos iteration, which converges quickly to the largest (not in magnitude) eigenvalues of a symmetric matrix, can be applied to A and $-A$ to quickly obtain both a feasible point and a subgradient at that point.

5.3.2 OSQP with an inexact quadratic solve

Consider the quadratic program

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^\top Px + q^\top x \\ \text{subject to} & l \le Ax \le u, \end{array}$$

where $x \in \mathbb{R}^n$ is the variable and $P = P^\top \succeq 0$, $q \in \mathbb{R}^n$, and $A \in \mathbb{R}^{m\times n}$ are data. Let $\mathcal{Z} = \{x \mid l \le x \le u\}$, where we allow some bounds to be infinite. We also assume that a primal-dual solution exists.

For $\tau, \rho > 0$ and $0 < \theta < 2$, OSQP uses the updates [Stellato et al., 2020, Algorithm 1]

$$\begin{aligned} \begin{bmatrix} P + \tau I & A^\top \\ A & -\rho^{-1}I \end{bmatrix} \begin{bmatrix} \tilde{x}_{k+1} \\ \nu_{k+1} \end{bmatrix} &= \begin{bmatrix} \tau x_k - q \\ z_k - \rho^{-1}y_k \end{bmatrix}, \\ \tilde{z}_{k+1} &= z_k + \rho^{-1}(\nu_{k+1} - y_k), \\ x_{k+1} &= \theta\tilde{x}_{k+1} + (1-\theta)x_k, \\ z_{k+1} &= \Pi_{\mathcal{Z}}\big(\theta\tilde{z}_{k+1} + (1-\theta)z_k + \rho^{-1}y_k\big), \\ y_{k+1} &= y_k + \rho\big(\theta\tilde{z}_{k+1} + (1-\theta)z_k - z_{k+1}\big). \end{aligned}$$

Here τ and θ denote OSQP's parameters σ and α, respectively. Let $K = \begin{bmatrix} P + \tau I & A^\top \\ A & -\rho^{-1} I \end{bmatrix}$ and fix τ, ρ, and θ.

Equip $\mathbb{R}^{n+m}$ with the inner product $\langle v, w \rangle_H = v^\top H w$, where $H = \mathbf{diag}(\tau I_n, \rho I_m)$. Define

$$\begin{aligned} f(x,z) &= \frac{1}{2} x^\top P x + q^\top x + \mathcal{I}_{\{(x,z):Ax=z\}}(x,z), & g(x,z) &= \mathcal{I}_{\mathcal{Z}}(z), \\ A_1 &= H^{-1} \partial f, & A_2 &= H^{-1} \partial g. \end{aligned} \tag{36}$$

Both operators are maximal monotone in this inner product. Write $F = J_{A_1}$ and $G = J_{A_2}$, so $G(x,z) = (x, \Pi_{\mathcal{Z}}(z))$. For $S_k = (x_k, z_k + \rho^{-1} y_k)$, the OSQP updates become

$$S_{k+1} = S_k + \theta\big(F(2G(S_k) - S_k) - G(S_k)\big). \tag{37}$$

This identity holds whenever $y_k \in N_{\mathcal{Z}}(z_k)$, which the projection enforces for $k \geq 1$. This is relaxed DRS with the operator order (A_2, A_1).

To make the quadratic solve inexact in Algorithm 2, we reverse this order:

$$W_{k+1} = T_\theta W_k, \qquad T_\theta = I + \theta\big(G(2F - I) - F\big). \tag{38}$$

Since F is affine, $S_k = (2F - I)W_k$ maps every exact trajectory of (38) to (37) with corresponding initialization [Bauschke and Moursi, 2016]. The inexact iterates below use this reordered state W_k.

Apply Algorithm 2 to the preconditioned subdifferential operators (36) with stepsize $\gamma = 1$, the H-norm, and the relaxed update of Proposition 4.1. At $W_k = (w_{x,k}, w_{z,k})$, the first resolvent is the quadratic subproblem

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}\tilde{x}^\top P \tilde{x} + q^\top \tilde{x} + \frac{\tau}{2}\|\tilde{x} - w_{x,k}\|^2 + \frac{\rho}{2}\|\tilde{z} - w_{z,k}\|^2 \\ \text{subject to} & A\tilde{x} = \tilde{z}, \end{array} \tag{39}$$

with variables $(\tilde{x}, \tilde{z})$ and Lagrange multiplier ν. For an approximate pair $(\tilde{x}, \nu)$, compute

$$h = P\tilde{x} + q + A^\top \nu, \qquad \bar{z} = \Pi_{\mathcal{Z}}(A\tilde{x} + \rho^{-1}\nu).$$

The two graph pairs in Algorithm 2 are (X, U) and (Y, V), where

$$\begin{aligned} X &= \begin{bmatrix} \tilde{x} \\ A\tilde{x} \end{bmatrix}, & U &= \begin{bmatrix} \tau^{-1} h \\ -\rho^{-1}\nu \end{bmatrix} \in A_1 X, \\ Y &= \begin{bmatrix} \tilde{x} - \tau^{-1} h \\ \bar{z} \end{bmatrix} = G(X - U), & V &= \begin{bmatrix} 0 \\ A\tilde{x} + \rho^{-1}\nu - \bar{z} \end{bmatrix} \in A_2 Y. \end{aligned} \tag{40}$$

The choice $X_z = A\tilde{x}$ ensures $X \in \mathbf{dom}\, A_1$ even before the quadratic solve converges. This corresponds to eliminating the varible $\tilde{z}$ from the quadratic subproblem (39). Set

$$\begin{aligned} \zeta &= X - Y = U + V = \begin{bmatrix} \tau^{-1} h \\ A\tilde{x} - \bar{z} \end{bmatrix}, \\ e &= X + U - W_k = \begin{bmatrix} \tilde{x} + \tau^{-1} h - w_{x,k} \\ A\tilde{x} - \rho^{-1}\nu - w_{z,k} \end{bmatrix}. \end{aligned} \tag{41}$$

The terminal criterion becomes stop if $\|\zeta\|_H \leq \delta$. Otherwise, we accept the candidate update when

$$\|e\|_H \leq \sigma \|e - \zeta\|_H,$$

for $0 < \sigma < \min\left\{1, \frac{2-\theta}{\theta}\right\}$. We then update

$$w_{x,k+1} = w_{x,k} - \frac{\theta}{\tau} h,$$
$$w_{z,k+1} = w_{z,k} - \theta(A\tilde{x} - \bar{z}).$$

If neither test passes, we run s additional iterations of the inner procedure and recompute the residuals (40) and (41). Moreover, we warm-start $(\tilde{x}, \nu)$ at the previously accepted value, which satisfies the warm-start hypotheses of Proposition 4.1.

GMRES for the quadratic subproblem. The KKT conditions of (39) reduce to

$$K \begin{bmatrix} \tilde{x} \\ \nu \end{bmatrix} = b_k := \begin{bmatrix} \tau w_{x,k} - q \\ w_{z,k} \end{bmatrix}.$$

The matrix is nonsingular because eliminating ν gives $P + \tau I + \rho A^\top A \succ 0$. For the true linear-system residual

$$r = \begin{bmatrix} r_x \\ r_z \end{bmatrix} := K \begin{bmatrix} \tilde{x} \\ \nu \end{bmatrix} - b_k,$$

the inexact DRS error satisfies

$$e = \begin{bmatrix} \tau^{-1} r_x \\ r_z \end{bmatrix}, \qquad \|e\|_H^2 = \tau^{-1}\|r_x\|^2 + \rho\|r_z\|^2.$$

All unweighted norms here are Euclidean.

To obtain contraction in this norm, apply GMRES [Saad and Schultz, 1986] to the rescaled system

$$\underbrace{\begin{bmatrix} I + \tau^{-1}P & \sqrt{\frac{\rho}{\tau}}\, A^\top \\ -\sqrt{\frac{\rho}{\tau}}\, A & I \end{bmatrix}}_{\widehat{K}} \begin{bmatrix} \sqrt{\tau}\,\tilde{x} \\ \rho^{-1/2}\nu \end{bmatrix} = \begin{bmatrix} \sqrt{\tau}\, w_{x,k} - \tau^{-1/2} q \\ -\sqrt{\rho}\, w_{z,k} \end{bmatrix}.$$

Its residual is $\widehat{r} = (\tau^{-1/2} r_x, -\sqrt{\rho}\, r_z)$, so $\|\widehat{r}\| = \|e\|_H$, and

$$\frac{\widehat{K} + \widehat{K}^\top}{2} = \mathbf{diag}(I + \tau^{-1}P, I) \succeq I.$$

Choose $\Lambda > 1$ with $\Lambda \geq \|\widehat{K}\|_2$. Any $\Lambda > 1 + \tau^{-1}\|P\|_2 + \sqrt{\frac{\rho}{\tau}}\|A\|_2$ suffices. At every GMRES iteration, the next search space contains a correction along the current residual. For a nonzero residual, minimizing over that direction gives

$$\|\widehat{r}^{j+1}\|^2 \leq \|\widehat{r}^j\|^2 - \frac{((\widehat{r}^j)^\top \widehat{K}\widehat{r}^j)^2}{\|\widehat{K}\widehat{r}^j\|^2} \leq (1 - \Lambda^{-2})\|\widehat{r}^j\|^2.$$

See Trefethen and Bau [1997] for more details. Thus, in exact arithmetic, full GMRES or restarted GMRES with any positive restart length satisfies

$$\|e^j\|_H^2 \leq \alpha^j \|e^0\|_H^2, \qquad C_0 = 1, \qquad \alpha = 1 - \Lambda^{-2}.$$

Choosing $\Lambda > 1 + \tau^{-1}\|P\|_2 + \sqrt{\frac{\rho}{\tau}}\|A\|_2$ gives $\alpha > 1 - (1 + \tau^{-1}\|P\|_2 + \sqrt{\frac{\rho}{\tau}}\|A\|_2)^{-2}$. With the stated warm start, Proposition 4.1 gives $O(\delta^{-2})$ total GMRES iterations for fixed admissible parameters.

Alternatively, eliminate ν and apply GMRES to the positive definite Schur complement system

$$Q\tilde{x} = c_k, \qquad Q := P + \tau I + \rho A^\top A, \qquad c_k := \tau w_{x,k} - q + \rho A^\top w_{z,k}.$$

Recovering $\nu = \rho(A\tilde{x} - w_{z,k})$ gives $r_z = 0$ and $\|e\|_H = \tau^{-1/2}\|Q\tilde{x} - c_k\|$. For $\kappa = \kappa_2(Q) > 1$, applying the Chebyshev polynomial estimate in the proof of Trefethen and Bau [1997, Theorem 38.5] to the GMRES residual bound of Saad and Schultz [1986, Proposition 4] gives, for unrestarted GMRES,

$$\|Q\tilde{x}^j - c_k\| \le 2\left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^j \|Q\tilde{x}^0 - c_k\|.$$

Since $(I + \tau^{-1}P)^2 \succeq I + \tau^{-1}P$ implies $\kappa \le \|\widehat{K}\|_2^2 \le \Lambda^2$, the squared-residual geometric factor satisfies

$$\alpha_{\mathrm{S}} := \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^2 \le \left(\frac{\Lambda - 1}{\Lambda + 1}\right)^2 < 1 - \Lambda^{-2} = \alpha.$$

This compares worst-case geometric factors with the larger system. When $\kappa = 1$, one GMRES step solves the system.

To retain the graph warm start, test the previously accepted $(\tilde{x}, \nu)$ before reconstructing ν in the Schur iterations. Since $Q\tilde{x}^0 - c_k = r_x^0 + \rho A^\top r_z^0$, the bound gives $C_0 = 4(1 + \frac{\rho}{\tau}\|A\|_2^2)$ and $\alpha = \alpha_{\mathrm{S}}$ in Assumption 4.1 when $\kappa > 1$. Thus the same total-work argument applies.

5.4 Inexact Davis–Yin

We can extend the Douglas–Rachford result to the more general Davis–Yin case [Davis and Yin, 2017], where we wish to solve

$$\text{find } x \text{ such that } 0 \in A_1 x + A_2 x + A_3 x, \tag{42}$$

where A_1 and A_2 are maximal monotone operators and A_3 is L_3^{-1}-cocoercive.[3] Let $\tau \in (0, 2/L_3)$ be the Davis–Yin stepsize and let

$$0 < \theta \le 1 - \frac{\tau L_3}{4}$$

be the relaxation parameter. As in the Douglas–Rachford case, the stepsize of Algorithm 1 is $\gamma = 1$, and τ plays the role of the stepsize inside A. We define

$$A = \begin{bmatrix} \tau A_1 & I & -I \\ \tau A_3 - I & \tau A_2 & I \\ \theta I & -\theta I & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} I \\ -I \\ I \end{bmatrix}, \qquad \gamma = 1. \tag{43}$$

If $A_3 = 0$, one may take $\theta = 1$ (formally $L_3 = 0$), and the operators (43) reduce to the Douglas–Rachford operators above. Unlike in the Douglas–Rachford case, A is *not* monotone in general: its linear part is not skew-adjoint when $\theta \neq 1$ or $A_3 \neq 0$ (for instance, with $A_1 = A_2 = 0$ and $A_3 = L_3 I$ on $\mathbb{R}$, the symmetric part of A has zero diagonal and nonzero off-diagonal entries, so it is indefinite). Hence Assumption 2.1 does not hold. However, the operator B of Definition 2.1 is still maximal monotone (Proposition 5.4), so we may use the weaker Assumption 2.2, and the results in

Algorithm 4 Inexact Davis–Yin splitting as an instance of Algorithm 1.

1: **Initialize:** $w_0 \in \mathcal{H}$, stepsize $\tau \in \left(0, \frac{2}{L_3}\right)$, relaxation $\theta \in (0, 1 - \frac{\tau L_3}{4}]$, inexactness $\sigma \in [0, 1)$, stride $s \in \mathbb{N}$, and stopping tolerance δ
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Produce a pair $(x_k, u_k) \in \mathcal{H} \times \mathcal{H}$ with

$$u_k \in A_1 x_k, \qquad \tau u_k + x_k \approx w_k.$$

4: Compute $y_k = J_{\tau A_2}(x_k - \tau u_k - \tau A_3 x_k)$.
5: **if** $\theta\|x_k - y_k\| \le \delta$ **then**
6: Stop and return (x_k, y_k, u_k, w_k).
7: **else if** $\|x_k + \tau u_k - w_k\| \le \sigma\|(1-\theta)x_k + \theta y_k + \tau u_k - w_k\|$ **then**
8: Set $w_{k+1} = w_k - \theta(x_k - y_k)$.
9: **else**
10: Improve the pair (x_k, u_k) with $u_k \in A_1 x_k$ to reduce $\|x_k + \tau u_k - w_k\|$ via the iterative procedure s times and go to line 4.
11: **end if**
12: **end for**

Sections 3 and 4 still hold. Substituting this into Algorithm 1 yields Algorithm 4. We formalize this in Proposition 5.4.

In the notation of Algorithm 1, the quantities tested in lines 5 and 7 are $\|z_k\|$, $\|\gamma z_k + C^* u_k - w_k\|$, and $\|C^* u_k - w_k\|$, with $z_k = \theta(x_k - y_k)$ and $C^* u_k = (1-\theta)x_k + \theta y_k + \tau u_k$ (Proposition 5.4). Each check costs one evaluation of A_3 and one resolvent of A_2. When the A_1-step is exact, $\tau u_k = w_k - x_k$, and lines 4 and 8 become $y_k = J_{\tau A_2}(2x_k - w_k - \tau A_3 x_k)$ and $w_{k+1} = w_k + \theta(y_k - x_k)$, i.e., Davis–Yin splitting with relaxation parameter θ.

Proposition 5.4. *Let A_1 and A_2 be maximal monotone, let A_3 be L_3^{-1}-cocoercive, let $\tau \in (0, 2/L_3)$ and $\theta \in (0, 1 - \frac{\tau L_3}{4}]$, and let A and C be as in (43). The following hold:*

1. *C^* is surjective, and $0 \in A(x_1, x_2, v)$ if and only if $x_1 = x_2 =: x$, $v - x \in \tau A_1 x$, and $x - v - \tau A_3 x \in \tau A_2 x$. In particular, $\mathbf{zer}\, A \neq \varnothing$ if and only if (42) has a solution.*

2. *$(C^*(x_1, x_2, v), \zeta) \in \mathbf{gra}\, B$ if and only if there is $u \in A_1 x_1$ with*

$$x_2 = J_{\tau A_2}(x_1 - \tau u - \tau A_3 x_1), \qquad v = \tau u - \theta x_1 + (1+\theta)x_2, \qquad \zeta = \theta(x_1 - x_2). \tag{44}$$

   In that case, $C^*(x_1, x_2, v) = (1-\theta)x_1 + \theta x_2 + \tau u$, and $\zeta + C^*(x_1, x_2, v) - w = x_1 + \tau u - w$ for every w. Consequently, Algorithm 4 is Algorithm 1 applied to the operators (43) with (x_k, y_k, u_k) in the roles of (x_1, x_2, u): lines 5, 7, and 8 are its terminal test, relative error test, and update, and the warm start of Lemma 4.1 is the warm start of the A_1-solver at (x_{k-1}, u_{k-1}).

3. *B is maximal monotone, and $J = (I + B)^{-1}$ is the relaxed Davis–Yin map*

$$J(w) = w - \theta(x - y), \qquad x = J_{\tau A_1}(w), \quad y = J_{\tau A_2}(2x - w - \tau A_3 x).$$

[3] That is, $\langle A_3 x - A_3 x', x - x' \rangle \ge L_3^{-1}\|A_3 x - A_3 x'\|^2$ for all x and x' (for example $A_3 = \nabla h$ with h convex and L_3-smooth).

4. At termination of Algorithm 4, $u_k \in A_1 x_k$ and $v_k := \frac{1}{\tau}(x_k - y_k) - u_k - A_3 x_k \in A_2 y_k$, with

$$\|x_k - y_k\| \le \frac{\delta}{\theta}, \qquad \|u_k + v_k + A_3 x_k\| = \frac{1}{\tau}\|x_k - y_k\| \le \frac{\delta}{\tau\theta}.$$

The proof of Proposition 5.4 is in Appendix G.

Remark 5.2. Item 3 is consistent with the fact that Davis–Yin splitting is $\frac{2}{4-\tau L_3}$-averaged for $\tau \in (0, \frac{2}{L_3})$ [Davis and Yin, 2017, Proposition 2.1]. Relaxing it by $\theta = 1 - \frac{\tau L_3}{4}$ gives a $\frac{1}{2}$-averaged, i.e., firmly nonexpansive, map, which is the resolvent of a maximal monotone operator. The computation in the proof only uses $0 < \theta \le 1 - \frac{\tau L_3}{4}$, so it shows that B is maximal monotone for every $\tau L_3 < 4$, not only for $\tau < \frac{2}{L_3}$.

Remark 5.3. We also note that the A defined in (43) is not the same as that of Alves et al. [2026, Section 3.3]. Our construction highlights the expressive power of the framework, where the underlying operator need not necessarily be maximal monotone, as long as one can find the appropriate C to make Assumption 2.2 hold.

The ability to represent Davis–Yin splitting is of particular note, since this implies that the framework of Bredies et al. [2022] and Alves et al. [2026] encompasses Douglas–Rachford and its variants, as well as forward-backward splitting. This means that this framework encompasses ADMM, Chambolle–Pock, and proximal gradient descent, among others.

TODO: redo numerical experiments

6 Conclusion

We analyzed the expected inner work of proximal point methods for solving a monotone inclusion problem in a Hilbert space. Equipped with a contractive inner procedure for computing the resolvent, we provide a quantitative bound on the amount of expected inner work required to solve the monotone inclusion problem to a given accuracy. Our framework encompasses a broad variety of popular operator splitting algorithms, and we provide principled heuristics for selecting algorithm parameters a priori. Numerical experiments validate our findings and suggest that in practice, many algorithms can tolerate significant amounts of inexactness.

Many questions remain open. The $\mathbf{E}[K_\delta]$ term in Theorem 4.2 opens the door to considering average-case complexity, rather than the worst-case performance. Moreover, the choice of problem parameters is not only problem-sensitive, but also sensitive to the choice of inner procedure used. Providing more thorough guidelines on the selection of parameters would be useful in practice. Also of interest is differentiating which inner solvers are more compatible with certain outer solvers, and providing guidance on when to use a particular splitting or method over another. Another promising direction is an adaptive choice of parameters. The analysis of this work assumes that algorithm parameters are fixed, but in practice this is undesirable. Incorporating an analysis of adaptive parameter selection would bring the theory closer to practice. Another direction would be to relax the requirement in Algorithm 1 that $Cz_k \in Au_k$, for cases where finding a member of Au_k is difficult, such as the case where the proximal step is projection onto a convex set.

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A Tightness of Bounds

Here, we consider the tightness of the bounds of Section 4.

A.1 Dependence on the Stopping Tolerance

The additive logarithm in Theorem 4.2 and the worst-case rate in Theorem 3.1 describe different regimes. The following two constructions establish their respective dependence on δ.

Proposition A.1 (Necessity of the additive logarithm)**.** *Fix $\sigma \in (0,1)$, $\alpha \in (0,1)$, and a positive integer s. There is a scalar strongly convex quadratic, with $\gamma = 1$, a fixed initialization, and gradient descent as the inner procedure, satisfying Assumption 4.1 with $C_0 = 1$, for which every $\delta \in (0,1)$ gives*

$$K_\delta = 1, \qquad \sum_{k=0}^{K_\delta} N_k = s \left\lceil \frac{\log(1/\delta^2)}{s \log(1/\alpha)} \right\rceil. \tag{45}$$

The right-hand side of (13) in Theorem 4.2 exceeds this work by an additive term not dependent on δ.

Proof. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}$, $C = I$, $\gamma = 1$, and $A = B = \nabla f$, where

$$f(y) = \frac{\mu}{2} y^2, \qquad \mu = 1 + \frac{1}{\sigma}.$$

Set $w_0 = 1$ and initialize the first graph pair at $y^0 = \frac{1}{\mu}$ and $z^0 = 1$. At every outer iteration, apply gradient descent to $f(y) + \frac{1}{2}(y - w_k)^2$ with stepsize $(1 - \sqrt{\alpha})/(1 + \mu)$, using the graph pair $z^l = \mu y^l$. By Nesterov [2013, Theorem 2.1.15], the update and residual satisfy

$$y^{l+1} = y^l - \frac{1 - \sqrt{\alpha}}{1 + \mu} \left((1 + \mu) y^l - w_k \right),$$
$$e^{l+1} = \sqrt{\alpha}\, e^l, \qquad e^l = (1 + \mu) y^l - w_k.$$

Hence the deterministic contraction holds with equality, with the same α and $C_0 = 1$ at every outer iteration.

At the first check, the terminal test fails because $z^0 = 1 > \delta$, whereas the relative error test holds with equality:

$$|z^0 + y^0 - w_0| = \frac{1}{\mu} = \sigma \left(1 - \frac{1}{\mu} \right) = \sigma |y^0 - w_0|.$$

Thus $N_0 = 0$, the accepted pair is $(y_0, z_0) = (\frac{1}{\mu}, 1)$, and $w_1 = w_0 - z_0 = 0$. Warm-starting at this pair gives, during the second inner solve,

$$y^l = \frac{\alpha^{l/2}}{\mu}, \qquad z^l = \alpha^{l/2}, \qquad |e^l| = \left(1 + \frac{1}{\mu}\right)\alpha^{l/2} > \frac{\sigma}{\mu}\alpha^{l/2} = \sigma|y^l - w_1|.$$

The relative error test therefore fails at every finite inner iteration. The terminal test first passes at the first multiple l of s with $\alpha^{l/2} \le \delta$, proving (45).

For this choice of initialization, $R(w_0) = \frac{\mu}{1+\mu} = \frac{\sigma+1}{2\sigma+1}$ and

$$\frac{(1+\sigma)^2}{\sigma^2}|z^0 + y^0 - w_0|^2 = 1.$$

Consequently, the two logarithmic initialization terms in (13) have

$$\begin{aligned}
(\log \beta_0)_+ + \log \frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2} &= \left(\log \frac{1}{\max\left\{\left(\frac{\mu}{1+\mu}\right)^2, \delta^2\right\}}\right)_+ + \log \frac{\max\left\{\left(\frac{\mu}{1+\mu}\right)^2, \delta^2\right\}}{\delta^2} \\
&= \left(\log \frac{1}{\max\left\{\left(\frac{\sigma+1}{2\sigma+1}\right)^2, \delta^2\right\}}\right)_+ + \log \frac{\max\left\{\left(\frac{\sigma+1}{2\sigma+1}\right)^2, \delta^2\right\}}{\delta^2} \\
&= \log(\delta^{-2}),
\end{aligned}$$

where we used the fact that $\max\left\{\left(\frac{\sigma+1}{2\sigma+1}\right)^2, \delta^2\right\} < 1$. Since $K_\delta = 1$, the right-hand side of the total inner work bound (13) of Theorem 4.2 is

$$\frac{2s}{1-\alpha^s} + \frac{\log D_\sigma}{\log(1/\alpha)} + \frac{\log(\delta^{-2})}{\log(1/\alpha)}.$$

Noting that the ceiling in (45) adds fewer than s iterations to the last term completes the proof. □

Remark A.1 (Any prescribed positive number of outer updates). The construction of Proposition A.1 extends to any prescribed positive integer m if the gradient descent stepsize may depend on the outer state. Fix $\sigma \in (0,1)$, $\alpha \in (0,1)$, and a positive integer s, and choose

$$\mu \ge \max\left\{1 + \frac{1}{\sigma}, \alpha^{-s/4}\right\}, \qquad q := \sqrt{\alpha}, \qquad \rho := \mu^{-2/s} \le q.$$

Use the same quadratic $f(y) = \frac{\mu}{2}y^2$, with $C = I$ and $\gamma = 1$, for every m. At outer state w, apply gradient descent to $f(y) + \frac{1}{2}(y-w)^2$ with the stepsize

$$\eta(w) = \frac{1}{1+\mu}\begin{cases} 1, & w > 1, \\ 1-\rho, & w = 1, \\ 1-q, & w < 1. \end{cases}$$

This rule is independent of m and δ. Every graph residual contracts in one inner iteration by a factor of 0, ρ, or q, respectively, so Assumption 4.1 holds with the prescribed α and $C_0 = 1$.

For $m \geq 2$, set $w_0 = (1+\mu)^{m-1}$ and initialize the first graph pair at $y^0 = \frac{w_0}{1+\mu}$ and $z^0 = \mu y^0$. For $m = 1$, set $w_0 = 1$, $y^0 = \frac{1}{\mu}$, and $z^0 = 1$. For every $\delta \in (0,1)$, this gives

$$K_\delta = m, \qquad \sum_{k=0}^{K_\delta} N_k = s(m-1) + s\left\lceil \frac{\log(1/\delta^2)}{s\log(1/\alpha)} \right\rceil. \tag{46}$$

To verify the claim for $m \geq 2$, note that the first $m-1$ accepted updates are exact proximal steps, with

$$w_k = (1+\mu)^{m-1-k} \quad (0 \leq k \leq m-1).$$

The initial pair passes the relative error test without inner work. At each subsequent warm start before $w = 1$, the previous exact pair has $y^0 = w_k$ and $z^0 = \mu w_k > \delta$, so both tests fail. The inner solver reaches the exact pair after one iteration, which is checked after s iterations. At $w_{m-1} = 1$, the warm start is $(y^0, z^0) = (1, \mu)$, with residual $e^0 = \mu$. After s inner iterations, $e^s = \rho^s \mu = \frac{1}{\mu}$, giving the graph pair $(y^s, z^s) = (\frac{1}{\mu}, 1)$. This pair passes the relative error test because $\frac{1}{\mu} \leq \sigma(1 - \frac{1}{\mu})$, and its accepted update gives $w_m = 0$. All preceding accepted operator values are at least 1, so none passes the terminal test for $\delta < 1$. The final warm-started inner solve has $z^l = q^l$ and never passes the relative error test, exactly as in Proposition A.1. This proves (46), including $m = 1$ with its specified initialization.

The actual work and the upper bound in (13) are both $\Theta(m + \log(1/\delta))$, with constants depending only on the fixed parameters. Indeed, for $m \geq 2$ the cold start is exact, so $\beta_0 = 0$, while $R(w_0) = \mu(1+\mu)^{m-2}$. The endpoint logarithm is therefore $2\log\mu + 2(m-2)\log(1+\mu) + \log(1/\delta^2)$. In this family, the initial distance $(1+\mu)^{m-1}$ grows with m. Alternatively, rescaling every iterate and the stepsize rule's threshold by $h_m = (1+\mu)^{-(m-1)}$ gives initial distance 1, with $K_\delta = m$ for $0 < \delta < h_m$ and total work

$$s(m-1) + s\left\lceil \frac{\log(h_m^2/\delta^2)}{s\log(1/\alpha)} \right\rceil.$$

Proposition A.2 (Matching worst-case accuracy bounds)**.** *Fix $\sigma \in (0,1)$ and $\alpha \in (0,1)$, and set $\gamma = s = C_0 = 1$. For every $\delta \in (0, 1/2)$, there is an instance on $\mathbb{R}^2$ with $\mathbf{dist}(w_0, \mathbf{zer}\, B) = 1$ and an inner procedure satisfying Assumption 4.1 for which*

$$\sum_{k=0}^{K_\delta} N_k = K_\delta + 1 = \left\lceil \frac{\log 4}{\log(1+4\delta^2)} \right\rceil \geq \frac{\log 4}{4\delta^2}. \tag{47}$$

The upper bound in (13) is $O(\delta^{-2})$ on this family, with a constant independent of δ.

Proof. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}^2$, $C = I$, $w_0 = (1, 0)$, and

$$A = B = 2\delta U, \qquad U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The operator is monotone because U is skew-symmetric. Since $I + 2\delta U$ is invertible, Minty's Theorem shows that U is maximal monotone [Bauschke and Combettes, 2017, Theorem 21.1]. Its unique zero is the origin, and

$$J = \frac{I - 2\delta U}{1 + 4\delta^2}, \qquad J^\top J = \frac{I}{1+4\delta^2}.$$

Initialize the first graph pair at $y^0 = w_0$ and $z^0 = 2\delta U w_0$. Choose the inner procedure that returns $(J(w_k), 2\delta U J(w_k))$ after one inner iteration and retains this exact pair at all subsequent inner iterations. This procedure satisfies the deterministic contraction with $C_0 = 1$ and any prescribed $\alpha \in (0, 1)$, since its residual vanishes after the first iteration. The displayed formula for J evaluates this pair with a fixed number of scalar arithmetic operations.

At the initial check, $y^0 = w_0$ and $\|z^0\| = 2\delta > \delta$, so both tests fail. At every subsequent warm start, $y^0 = y_{k-1} = w_k$, while $z^0 = z_{k-1}$ has norm greater than δ because the preceding update was nonterminal. Thus both tests again fail before the first inner iteration, and $N_k = 1$ for every $0 \le k \le K_\delta$. All accepted updates are exact proximal steps, so

$$w_k = J^k w_0, \qquad y_k = J^{k+1} w_0, \qquad \|z_k\| = \frac{t}{(1+4\delta^2)^{(k+1)/2}}.$$

The first terminal check is therefore characterized by

$$K_\delta + 1 = \left\lceil \frac{\log(\frac{4\delta^2}{\delta^2})}{\log(1+4\delta^2)} \right\rceil = \left\lceil \frac{\log 4}{\log(1+4\delta^2)} \right\rceil.$$

The inequality $\log(1+x) \le x$ proves the lower bound in (47). Conversely, $\log(1+x) \ge \frac{x}{1+x}$ gives

$$K_\delta + 1 \le \frac{\log 4}{4\delta^2} + \log 4 + 1.$$

Moreover,

$$\frac{\|R(w_0)\|^2}{\delta^2} = \frac{4}{1+4\delta^2}, \qquad \beta_0 = \frac{(1+\sigma)^2}{\sigma^2}(1+4\delta^2).$$

Both quantities are bounded independently of δ, so (13) is also $O(\delta^{-2})$ on this family. □

Remark A.2 (Two subspaces). The resolvent in Proposition A.2 is also a Douglas–Rachford map for two lines. Let P_1 and P_2 be the orthogonal projectors onto $\{(a, 0) \mid a \in \mathbb{R}\}$ and $\{(a, -2\delta a) \mid a \in \mathbb{R}\}$, respectively. Direct multiplication gives

$$I - P_1 + P_2(2P_1 - I) = \frac{1}{1+4\delta^2} \begin{bmatrix} 1 & 2\delta \\ -2\delta & 1 \end{bmatrix} = J.$$

Thus the same trajectory and stopping residual occur for exact Douglas–Rachford splitting between these subspaces.

B Proof of Lemma 4.1

B.1 Proof of Bounds G_σ and H_σ

Proof. From the update rule of Algorithm 1,

$$\begin{aligned} \gamma z_k + C^* u_k - w_{k+1} &= \gamma z_k + y_k - (w_k - \gamma z_k) \\ &= e_k + \gamma z_k. \end{aligned}$$

Write $r := \|R(w_k)\|$, $e_J := y_k - J(w_k)$, $e_R := \gamma z_k - R(w_k)$, and $\varepsilon := \|e_k\|$, so that $e_k = e_J + e_R$, $\gamma z_k = R(w_k) + e_R$, $y_k - w_k = e_J - R(w_k)$, and

$$\begin{aligned} \gamma z_k + C^* u_k - w_{k+1} &= e_k + \gamma z_k \\ &= R(w_k) + e_J + 2e_R. \end{aligned}$$

We use three facts. First, (5) in Lemma 3.1 gives $\|e_J\|^2 + \|e_R\|^2 \le \varepsilon^2$. In particular, $\langle e_J, e_R\rangle \ge 0$, so $\|e_R\| \le \varepsilon$ and $\|e_R - e_J\|^2 = \varepsilon^2 - 4\langle e_J, e_R\rangle \le \varepsilon^2$. Second, the relative error test gives $\|R(w_k) - e_J\| = \|y_k - w_k\| \ge \frac{\varepsilon}{\sigma}$. Third, by the bound $\|\gamma z + y - w\| \le \frac{\sigma}{1-\sigma}\|R(w)\|$ (8) in Lemma 3.1, $\varepsilon \le \frac{\sigma r}{1-\sigma}$.

If $\varepsilon \le \sigma r$, then

$$\|R(w_k) + e_J + 2e_R\| \le r + \|e_J + e_R\| + \|e_R\| \le r + 2\varepsilon \le (1 + 2\sigma)r \le G_\sigma r.$$

Suppose now $\varepsilon \ge \sigma r$. Expanding, and using $\|R(w_k) + e_J\|^2 = 2r^2 + 2\|e_J\|^2 - \|R(w_k) - e_J\|^2$ and $4\langle e_J, e_R\rangle + 4\|e_R\|^2 = 4\varepsilon^2 - 4\|e_J\|^2 - 4\langle e_J, e_R\rangle$,

$$\begin{aligned}\|R(w_k) + e_J + 2e_R\|^2 &= \|R(w_k) + e_J\|^2 + 4\langle R(w_k), e_R\rangle + 4\langle e_J, e_R\rangle + 4\|e_R\|^2 \\ &= 2r^2 - 2\|e_J\|^2 - \|R(w_k) - e_J\|^2 + 4\varepsilon^2 + 4\langle R(w_k), e_R\rangle - 4\langle e_J, e_R\rangle.\end{aligned}$$

Since $\langle e_J, e_R\rangle \ge 0$, we have $\|R(w_k) - e_J\|^2 \ge \frac{\varepsilon^2}{\sigma^2}$, and

$$\langle R(w_k), e_R\rangle = \langle R(w_k), e_J\rangle + \langle R(w_k), e_R - e_J\rangle \le \langle R(w_k), e_J\rangle + r\varepsilon.$$

Moreover,

$$2\langle R(w_k), e_J\rangle = r^2 + \|e_J\|^2 - \|R(w_k) - e_J\|^2 \le r^2 + \|e_J\|^2 - \frac{\varepsilon^2}{\sigma^2}.$$

Therefore

$$\|R(w_k) + e_J + 2e_R\|^2 \le 4r^2 + 4r\varepsilon + \left(4 - \frac{3}{\sigma^2}\right)\varepsilon^2 =: \Phi(\varepsilon), \tag{48}$$

a quadratic in ε alone. Maximizing Φ over $\varepsilon \in [\sigma r, \frac{\sigma r}{1-\sigma}]$ yields (12).

We now move on to H_σ. Note that $w_{k+1} = w_k - \gamma z_k = J(w_k) - e_R$. Since R is firmly nonexpansive (Lemma 3.1) and $w_{k+1} - w_k = -\gamma z_k = -(R(w_k) + e_R)$,

$$\|R(w_{k+1}) - R(w_k)\|^2 \le \langle R(w_{k+1}) - R(w_k), w_{k+1} - w_k\rangle = -\langle R(w_{k+1}) - R(w_k), R(w_k) + e_R\rangle.$$

As a consequence,

$$\|R(w_{k+1})\|^2 \le \langle R(w_{k+1}), R(w_k) - e_R\rangle + \langle R(w_k), e_R\rangle.$$

The fact that $\langle R(w_{k+1}), R(w_k) - e_R\rangle \le \frac{1}{2}\|R(w_{k+1})\|^2 + \frac{1}{2}\|R(w_k) - e_R\|^2$ and using $\frac{1}{2}\|R(w_k) - e_R\|^2 + \langle R(w_k), e_R\rangle = \frac{1}{2}\left(r^2 + \|e_R\|^2\right)$ gives

$$\|R(w_{k+1})\|^2 \le r^2 + \|e_R\|^2 = \|R(w_k)\|^2 + \|\gamma z_k - R(w_k)\|^2. \tag{49}$$

The first two facts give $\|e_R\|^2 \le \varepsilon^2 - \|e_J\|^2 \le \sigma^2\|R(w_k) - e_J\|^2 - \|e_J\|^2 \le \sigma^2(\|e_J\| + r)^2 - \|e_J\|^2$, a concave quadratic in $\|e_J\|$ whose maximum, attained at $\|e_J\| = \frac{\sigma^2 r}{1-\sigma^2}$, equals $\frac{\sigma^2 r^2}{1-\sigma^2}$. Hence $\|e_R\| \le \frac{\sigma r}{\sqrt{1-\sigma^2}}$ and $\|R(w_{k+1})\|^2 \le r^2 + \frac{\sigma^2 r^2}{1-\sigma^2} = \frac{r^2}{1-\sigma^2} = H_\sigma^2 r^2$.

The proof of the tightness of the constants can be found in Appendix B.2. □

B.2 Tightness of G_σ and H_σ

Proof. The constant G_σ in (11a) is sharp. Fix σ and $r > 0$, let $\varepsilon \in [\sigma r, \frac{\sigma r}{1-\sigma}]$ maximize Φ in (48), and set, for an orthonormal pair (v_1, v_2),

$$\begin{aligned} e_J &= \|e_J\| v_1, \\ e_R &= \|e_R\| v_2, \\ \gamma\bar{z} &= \frac{r}{\varepsilon}(-\|e_J\| v_1 + \|e_R\| v_2), \end{aligned}$$

which is possible because $\|e_J\|^2 \geq 0$ for $\varepsilon \geq \sigma r$ and $\|e_R\|^2 \geq 0$ for $\varepsilon \leq \frac{\sigma r}{1-\sigma}$. This definition is not circular since we have

$$\begin{aligned} \|e_J\|^2 &= \frac{\varepsilon(\varepsilon^2 - \sigma^2 r^2)}{\sigma^2(\varepsilon + 2r)}, \\ \|e_R\|^2 &= \varepsilon^2 - \|e_J\|^2. \end{aligned}$$

Moreover, $\|\gamma\bar{z}\| = r$, $\langle e_J, e_R\rangle = 0$, $\|e_J + e_R\| = \varepsilon = \sigma\|e_J - \gamma\bar{z}\|$, and $\langle \gamma\bar{z}, e_R - e_J\rangle = r\|e_R - e_J\|$, so every inequality in the proof holds with equality and $\|\gamma\bar{z} + e_J + 2e_R\|^2 = \Phi(\varepsilon) = G_\sigma^2 r^2$. Such a configuration is realized by an accepted pair: for any $\bar{y}$, the two-point set $\{(\bar{y}, \bar{z}), (\bar{y}+e_J, \bar{z}+e_R/\gamma)\}$ is monotone and therefore extends to a maximal monotone operator B [Bauschke and Combettes, 2017, Theorem 20.21], for which $\bar{y} = J(w)$ and $\gamma\bar{z} = R(w)$ at $w = \bar{y} + \gamma\bar{z}$, and $(y, z) = (\bar{y} + e_J, \bar{z} + e_R/\gamma)$ is a graph pair at w with residual $e = e_J + e_R$ that passes the relative error test with equality. For $\sigma \leq \sigma_1$ the extremal configuration is $e_J = 0$, $e_R = \sigma R(w)$: the inner solver returns the exact resolvent point $y = J(w)$, but with $\gamma z = (1 + \sigma)R(w)$. For $\sigma \geq \sigma_2$ it is $e_R = 0$, $e_J = -\frac{\sigma}{1-\sigma}R(w)$.

The constant H_σ in (11b) is sharp as well, and here we can construct an explicit operator. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}^2$, $C = I$, and $A = B = \partial f$ with $f(y) = \frac{r}{\gamma}(|y_1| + \sigma H_\sigma |y_2|)$, a weighted ℓ_1 norm, so that J is the soft-thresholding map $J(w) = (\mathbf{sign}(w_1)\max\{|w_1| - r, 0\}, \mathbf{sign}(w_2)\max\{|w_2| - \sigma H_\sigma r, 0\})$. Fix $w_k = (w_1, 0)$ with $w_1 \geq \frac{2r}{1-\sigma^2}$, so that $J(w_k) = (w_1 - r, 0)$ and $R(w_k) = (r, 0)$, and let the accepted pair be

$$y_k = J(w_k) - \frac{\sigma^2}{1-\sigma^2}R(w_k) = \left(w_1 - \frac{r}{1-\sigma^2}, 0\right), \qquad \gamma z_k = (r, \sigma H_\sigma r),$$

a graph pair because $\gamma B(y_k) = \{r\} \times [-\sigma H_\sigma r, \sigma H_\sigma r]$ (as $y_{k,1} > 0$ and $y_{k,2} = 0$). In the notation of the proof, $e_J = -\frac{\sigma^2}{1-\sigma^2}R(w_k)$ and $e_R = (0, \sigma H_\sigma r) \perp e_J$, so $\|e_R\| = \frac{\sigma r}{\sqrt{1-\sigma^2}}$ attains the bound on $\|e_R\|$, and the residual $e_k = e_J + e_R$ satisfies $\|e_k\|^2 = \frac{\sigma^4 r^2}{(1-\sigma^2)^2} + \frac{\sigma^2 r^2}{1-\sigma^2} = \frac{\sigma^2 r^2}{(1-\sigma^2)^2} = \sigma^2\|y_k - w_k\|^2$: the relative error test passes with equality (and the terminal test fails whenever $\gamma\delta < \|\gamma z_k\| = \frac{r}{\sqrt{1-\sigma^2}}$). Then $w_{k+1} = w_k - \gamma z_k = (w_1 - r, -\sigma H_\sigma r)$ with $w_1 - r > r$, so $J(w_{k+1}) = (w_1 - 2r, 0)$, $R(w_{k+1}) = (r, -\sigma H_\sigma r) = R(w_k) - e_R$, and

$$\|R(w_{k+1})\|^2 = r^2 + \sigma^2 H_\sigma^2 r^2 = \|R(w_k)\|^2 + \|e_R\|^2 = \frac{r^2}{1-\sigma^2} = H_\sigma^2\|R(w_k)\|^2,$$

so both inequalities leading to (49) hold with equality. The extremal configurations for the two constants differ: for G_σ (with $\sigma \leq \sigma_1$) the error e_R in γz_k is aligned with $R(w_k)$, whereas for H_σ it is orthogonal to $R(w_k)$ and as large as the relative error test allows, since (49) adds it in quadrature. The first coordinate of this example is the instance of Remark B.1. □

Remark B.1 (Sharpness of Theorems 4.1, 4.2 and 4.4). The per-step bound (18) is attained, for every $\sigma \in (0,1)$, s, α, and C_0, by the following instance. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}$, $C = I$, $\gamma = 1$, and $A = B = \partial(r|\cdot|)$ with $r > \delta$, so that $J(w) = w - r$ and $R(w) = r$ for $w > r$, and let $w_0 > r$. At an outer iteration with $w_k > r$, so that $\|R(w_k)\| = r$, let the inner procedure return

$$\begin{aligned} y^l &= J(w_k) + t_l R(w_k), \\ z^l &= \frac{1}{\gamma} R(w_k), \\ t_l &:= \sqrt{C_0}\, \alpha^{l/2} t_0 \quad (l \geq 1), \\ t_0 &:= \frac{1}{r} \|\gamma z^0 + y^0 - w_k\|, \end{aligned}$$

that is, y^l moves along the segment from $J(w_k)$ toward w_k while the operator value stays exact. Then $y^l > 0$, so $(y^l, z^l) \in \mathbf{gra}\, B$, and the residual is $e^l = t_l R(w_k)$. Thus, $\|e^l\|^2 = C_0 \alpha^l \|e^0\|^2$ for $l \geq 1$. The relative error test $\|e^l\| \leq \sigma \|y^l - w_k\| = \sigma |1 - t_l| r$ holds if and only if $t_l \leq \frac{\sigma}{1+\sigma}$; i.e., if and only if $\|e^l\| \leq \frac{\sigma}{1+\sigma} \|R(w_k)\|$; and the terminal test fails because $\|z^l\| = r > \delta$. Hence the inner loop exits at the first check at which the sufficient condition used in the proof of Theorem 4.1 holds, and $N_k = s \left\lceil \frac{(\log \tilde{\beta}_k)_+}{s \log(1/\alpha)} \right\rceil$ exactly (whenever $\|e^0\| > \frac{\sigma}{1+\sigma} \|R(w_k)\|$, which is the case for $k \geq 1$ below). Since z is exact, the outer step is the exact proximal step $w_{k+1} = J(w_k)$, and the next warm start is $e^0_{k+1} = e_k + \gamma z_k = (1 + t_{\text{exit}}) R(w_k)$ with $t_{\text{exit}} \leq \frac{\sigma}{1+\sigma}$, so that $\|e^0_{k+1}\| \leq \frac{1+2\sigma}{1+\sigma} \|R(w_k)\|$ and, for $k \geq 1$,

$$\tilde{\beta}_k \leq \frac{(1+2\sigma)^2 C_0}{\sigma^2} \frac{\|R(w_{k-1})\|^2}{\|R(w_k)\|^2} = \frac{D}{(1+\sigma)^2} \frac{\|R(w_{k-1})\|^2}{\|R(w_k)\|^2} \qquad (\sigma \leq \sigma_1),$$

with equality in the inequality when $t_{\text{exit}} = \frac{\sigma}{1+\sigma}$, which is approached for $s = 1$ as $\alpha \to 1$. A randomized variant of the same instance, in which the residual decreases deterministically to just above the threshold $\frac{\sigma}{1+\sigma} \|R(w_k)\|$ and then, at each inner iteration, is either solved exactly with probability $1 - \alpha$ or left unchanged, makes Markov's inequality in the proof of Theorem 4.1 tight at every check. It shows that the bound on $\mathbf{E}_k[N_k]$ there, and with it the per-iteration constant $\frac{s}{1-\alpha^s}$ of Theorem 4.2, can be approached arbitrarily closely.

C Proof of the Relaxed Inexact PPM Proposition

Proof of Proposition 4.1. For an accepted graph pair (y, z) at w, write $d = \gamma z$, $h = w - y$, and $e = d - h$. The relative test gives $\|e\| \leq \sigma \|h\|$ and $\|d\| \leq (1 + \sigma)\|h\|$. Hence

$$\begin{aligned} 2\langle h, d \rangle &= \|h\|^2 + \|d\|^2 - \|e\|^2 \\ &\geq (1 - \sigma^2)\|h\|^2 + \|d\|^2 \geq \frac{2}{1+\sigma} \|d\|^2. \end{aligned}$$

Monotonicity implies $\langle y - w^\star, d \rangle \geq 0$. Expanding $\|w - \theta d - w^\star\|^2$ proves the decrease estimate (20). Before termination, each accepted point has $\|z_k\| > \delta$, so telescoping this inequality gives the outer stopping time bound (21). Each inner loop is finite almost surely by Theorem 4.1, whose proof carries over. The terminal test carries over similarly.

Recall that $e_k = \gamma z_k + y_k - w_k$ for an accepted pair and $R = I - J$. At the next outer iteration, reusing this pair gives

$$e^0_{k+1} = e_k + \theta \gamma z_k.$$

The estimates in Lemma 3.1 and nonexpansiveness of R therefore imply

$$\|e_{k+1}^0\| \le G_{\sigma,\theta}\|R(w_k)\|$$
$$\|R(w_{k+1})\| \le H_{\sigma,\theta}\|R(w_k)\|.$$

Moreover,

$$\frac{1+\sigma}{\sigma}G_{\sigma,\theta} - H_{\sigma,\theta} = \frac{\theta + 2\sigma^2}{\sigma(1-\sigma)} > 0,$$

so $D_{\sigma,\theta} \ge H_{\sigma,\theta}^2$. For $r_k = \max\{\|R(w_k)\|, \gamma\delta\}$, these estimates give

$$\beta_k \le D_{\sigma,\theta}\frac{r_{k-1}^2}{r_k^2},$$
$$r_k \le H_{\sigma,\theta} r_{k-1},$$

and the same statements hold with $\tilde{\beta}_k$ in the deterministic case. Thus

$$(\log \beta_k)_+ \le \log D_{\sigma,\theta} + \log \frac{r_{k-1}^2}{r_k^2}.$$

The per-solve bounds are unchanged, and this inequality telescopes as in the proofs of Theorems 4.2 and 4.4. Combining the resulting bounds with the outer stoppping time bound (21) proves the total-work claim. □

D Proof for Accelerated Inner Solves

Proof of Proposition 5.2. Fix an outer iteration k, and let $x_k^\star$ be the unique minimizer of φ_k. As in the gradient descent case, φ_k is μ-strongly convex and L_φ-smooth, with $\mu = \frac{1}{\gamma}$ and $\frac{L_\varphi}{\mu} = \kappa$. The graph pair $(x^l, \nabla f(x^l))$ has residual $e^l = \gamma\nabla\varphi_k(x^l)$. The standard estimate for Nesterov's method with zero initial momentum gives

$$\varphi_k(x^l) - \varphi_k(x_k^\star) \le \left(1 - \frac{1}{\sqrt{\kappa}}\right)^l \left(\varphi_k(x^0) - \varphi_k(x_k^\star) + \frac{\mu}{2}\|x^0 - x_k^\star\|^2\right). \tag{50}$$

This estimate follows from the estimating sequence analysis of Nesterov [2013, Theorem 2.2.3], with initial curvature μ and constant acceleration parameter $\sqrt{\frac{\mu}{L_\varphi}}$. Strong convexity yields

$$\frac{\mu}{2}\|x^0 - x_k^\star\|^2 \le \varphi_k(x^0) - \varphi_k(x_k^\star) \le \frac{1}{2\mu}\|\nabla\varphi_k(x^0)\|^2.$$

Smoothness gives $\|\nabla\varphi_k(x^l)\|^2 \le 2L_\varphi(\varphi_k(x^l) - \varphi_k(x_k^\star))$. Combining these inequalities with (50), we obtain

$$\begin{aligned}\|e^l\|^2 &\le 2\gamma^2 L_\varphi\left(\varphi_k(x^l) - \varphi_k(x_k^\star)\right)\\ &\le \frac{2L_\varphi}{\mu}\left(1 - \frac{1}{\sqrt{\kappa}}\right)^l \gamma^2\|\nabla\varphi_k(x^0)\|^2\\ &= 2\kappa\left(1 - \frac{1}{\sqrt{\kappa}}\right)^l \|e^0\|^2.\end{aligned}$$

This deterministic residual bound holds for every inner iteration and every outer warm start, proving (24).

Substituting these constants into Theorems 4.2 and 4.4 gives the stated coefficient of δ^{-2}. The parameter-selection argument in the proof of Proposition 5.1 uses only that $C_0 \geq 1$ and that α and C_0 do not depend on σ. These properties hold here, so that argument gives uniqueness, the equation (23), and $\sigma^\star < (3/2 + \chi + \log(2\kappa))^{-1}$. The final inequality follows from $\chi \geq s \log(1/\alpha)$. The remaining terms have the same dependence on δ as in the gradient descent case. □

E Proof of the Douglas–Rachford Specialization

Verification of Algorithm 2. Let A and C be the operators in (26). For any reduced input w, set

$$x_1 = J_{\tau A_1}(w), \qquad x_2 = J_{\tau A_2}(2x_1 - w), \qquad v = w - 2x_1 + 2x_2.$$

Substituting gives $Cw \in (CC^* + A)(x_1, x_2, v)$, so Assumption 2.1 holds. The zero conditions give $x_1 = x_2 =: x$, $v - x \in \tau A_1 x$, and $x - v \in \tau A_2 x$, which imply $0 \in A_1 x + A_2 x$.

For the graph points computed in lines 3–4 of Algorithm 2, define

$$U_k = (x_k, y_k, \tau u_k + 2y_k - x_k), \qquad \zeta_k = x_k - y_k.$$

The exact second resolvent gives $\tau(u_k + v_k) = x_k - y_k$. Together with $u_k \in A_1 x_k$ and $v_k \in A_2 y_k$, this yields

$$C\zeta_k \in AU_k, \qquad C^* U_k = y_k + \tau u_k, \qquad \zeta_k + C^* U_k - w_k = x_k + \tau u_k - w_k.$$

Thus, lines 5 and 7 are the terminal and relative error tests of Algorithm 1, and line 8 is its update $w_{k+1} = w_k - \zeta_k$ with $\gamma = 1$. At termination, $\|x_k - y_k\| \leq \delta$ and $\|u_k + v_k\| = \frac{1}{\tau}\|x_k - y_k\| \leq \frac{\delta}{\tau}$, giving the stated certificate. Since the exact second resolvent makes (U_k, ζ_k) a function of (x_k, u_k) alone, reusing the previously accepted (x_k, u_k) also reproduces the previous graph pair, as required for the warm start in Lemma 4.1. □

F Proof of the ADMM Specialization

Proof of Proposition 5.3. The operator A_E in (28) can be written as

$$A_E = \begin{bmatrix} \frac{1}{c}\partial f & 0 & 0 \\ 0 & \frac{1}{c}\partial g & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & E^\top & -E^\top \\ -E & 0 & I \\ E & -I & 0 \end{bmatrix}.$$

The first term is the subdifferential of the convex function $(x, z, w) \mapsto f(x) + g(z)$ and the second term is skew-symmetric and therefore maximal monotone. By Bauschke and Combettes [2017, Corollary 25.5], A_E is maximal monotone. Its zeros satisfy

$$Ex = z, \qquad a + E^\top b = 0, \qquad v = z - \frac{b}{c}, \qquad a \in \partial f(x), \quad b \in \partial g(z).$$

Conversely, any KKT point (x, z, a, b) gives a zero by taking $v = z - \frac{1}{c}b$. Thus, $\mathbf{zer}\, A_E \neq \varnothing$ if and only if the composite optimization problem (27) has a solution.

It is clear from inspection that $C_{E,\rho}$ has full column rank. Moreover,

$$C_{E,\rho}C_{E,\rho}^* + A_E = \begin{bmatrix} E^\top E + \rho I + \frac{1}{c}\partial f & 0 & 0 \\ -2E & I + \frac{1}{c}\partial g & 0 \\ 2E & -2I & I \end{bmatrix}.$$

For any reduced input $W = (w, \sqrt{\rho}t)$ (or $W = w$ when $\rho = 0$), the inclusion $C_{E,\rho}W \in (C_{E,\rho}C_{E,\rho}^* + A_E)(x, z, v)$ therefore has the unique solution

$$\begin{aligned} x &= \operatorname*{argmin}_\xi \left\{ f(\xi) + \frac{c}{2}\|E\xi - w\|^2 + \frac{c\rho}{2}\|\xi - t\|^2 \right\}, \\ z &= \operatorname{prox}_{g/c}(2Ex - w), \\ v &= w - 2Ex + 2z. \end{aligned}$$

When $\rho = 0$, the proximal term in the first objective vanishes. The first subproblem solution x exists and is unique because its objective is proper, lower semicontinuous, and $c\rho$-strongly convex when $\rho > 0$, or $c\lambda_{\min}(E^\top E)$-strongly convex when $\rho = 0$. Recall that in the $\rho = 0$ case we assume that E has full column rank, and thus $E^\top E \succ 0$. This proves Assumption 2.1 with $\gamma = 1$, with no rank assumption on E needed when $\rho > 0$.

Suppose $\rho > 0$. We fix an outer iteration and suppress the indexing for brevity. Given graph points $a \in \partial f(x)$ and $b \in \partial g(z)$, define

$$\begin{aligned} d &= Ex - z, \qquad q = \frac{a + E^\top b}{c} - E^\top d, \\ u &= (x, z, z - \frac{1}{c}b), \qquad \zeta = \left(d, \frac{q}{\sqrt{\rho}}\right). \end{aligned}$$

Direct substitution gives

$$\begin{aligned} C_{E,\rho}\zeta &= \begin{pmatrix} E^\top d + q \\ -d \\ d \end{pmatrix} \\ &= \begin{pmatrix} \frac{a + E^\top b}{c} \\ -Ex + z \\ Ex - z \end{pmatrix} \\ &\in A_E u, \\ C_{E,\rho}^* u &= \begin{pmatrix} Ex - \frac{1}{c}b \\ \sqrt{\rho}x \end{pmatrix}. \end{aligned}$$

Conversely, suppose $C_{E,\rho}\zeta \in A_E(x, z, v)$. Expanding the bottom row shows that $\zeta_1 = Ex - z = d$. The second row gives $b = c(z - v) \in \partial g(z)$, and hence $v = z - \frac{1}{c}b$. The first row then gives $a \in \partial f(x)$ with

$$\sqrt{\rho}\zeta_2 = \frac{a + E^\top b}{c} - E^\top d = q.$$

Thus every graph pair has this form.

At outer state $W_k = (w_k, \sqrt{\rho}t_k)$, write $z = z_{k+1}$ and use the residuals in (32). Their definitions imply

$$\begin{aligned} r_{f,k} + E^\top r_{g,k} &= q_k + \rho(x_k - t_k), \\ Ex_k - \frac{1}{c}b_k - w_k &= -d_k - r_{g,k}. \end{aligned}$$

Consequently,

$$\zeta + C^*_{E,\rho} u - W_k = \begin{pmatrix} -r_{g,k} \\ \frac{r_{f,k}+E^\top r_{g,k}}{\sqrt{\rho}} \end{pmatrix},$$
$$C^*_{E,\rho} u - W_k = \begin{pmatrix} -d_k - r_{g,k} \\ \sqrt{\rho}(x_k - t_k) \end{pmatrix}.$$

Thus the quantities in (34) satisfy $Q_k = \|\zeta\|$, $E_k = \|\zeta + C^*_{E,\rho} u - W_k\|$, and $P_k = \|C^*_{E,\rho} u - W_k\|$. The terminal and relative error tests are exactly those of Algorithm 1. Its update $W_{k+1} = W_k - \zeta$ gives

$$w_{k+1} = w_k - Ex_k + z_{k+1},$$
$$t_{k+1} = t_k - \frac{q_k}{\rho} = x_k - \frac{r_{f,k} + E^\top r_{g,k}}{\rho}.$$

Defining $p_{k+1} = c(z_{k+1} - w_{k+1})$ preserves $w_{k+1} = z_{k+1} - \frac{1}{c} p_{k+1}$ and yields

$$p_{k+1} = c(Ex_k - w_k) = p_k + c(Ex_k - z_k).$$

These are the updates of Algorithm 3.

Now suppose $\rho = 0$. Proximal optimality of the iterates from (31) gives $b_k \in \partial g(z_{k+1})$. Moreover, $E^\top \ell_k = a_k$ implies

$$q_k = \frac{a_k + E^\top b_k}{c} - E^\top (Ex_k - z_{k+1}) = 0.$$

Setting $u = (x_k, z_{k+1}, z_{k+1} - \frac{1}{c} b_k)$ and $\zeta = d_k$, substitution gives

$$C_{E,0}\zeta = \begin{pmatrix} E^\top d_k \\ -d_k \\ d_k \end{pmatrix} \in A_E u, \qquad C^*_{E,0} u = z_{k+1} + \frac{1}{c}\ell_k.$$

Conversely, suppose $C_{E,0}\zeta \in A_E(x, z, v)$. The third row gives $\zeta = Ex - z =: d$, the second gives $b = c(z - v) \in \partial g(z)$, and the first gives $a = cE^\top d - E^\top b \in \partial f(x)$. Taking $\ell = cd - b$ then gives $E^\top \ell = a$ and $z = \mathrm{prox}_{g/c}(Ex - \frac{1}{c}\ell)$. Thus (31) parametrizes every graph pair for $\rho = 0$.

With $e_k = Ex_k + \frac{1}{c}\ell_k - w_k$, the residual definitions yield

$$r_{g,k} = -e_k, \qquad r_{f,k} = E^\top e_k, \qquad r_{f,k} + E^\top r_{g,k} = 0.$$

Consequently, the quantities in (35) satisfy

$$Q_k = \|\zeta\|, \qquad E_k = \|\zeta + C^*_{E,0} u - w_k\|, \qquad P_k = \|C^*_{E,0} u - w_k\|.$$

These are precisely the abstract terminal and relative error tests.

For $\rho > 0$, the two residuals are scaled subgradients of the objectives in (30). In particular, $r_{f,k} = 0$ gives

$$0 \in \partial f(x_k) + E^\top p_k + cE^\top (Ex_k - z_k) + c\rho(x_k - t_k).$$

After the multiplier update, $r_{g,k} = 0$ gives

$$0 \in \partial g(z_{k+1}) - p_{k+1} + c(z_{k+1} - Ex_k).$$

When $\rho > 0$ and both solves are exact, the auxiliary update gives $t_{k+1} = x_k$. Hence the exact method is proximal ADMM, with the additional term $\frac{c\rho}{2}\|x - t_k\|^2$ in the first subproblem and the ordinary second subproblem. For $k \geq 1$, this proximal term is centered at the previously accepted x_{k-1}. When $\rho = 0$, acceptance with $E_k = 0$ gives $\ell_k = c(w_k - Ex_k) = -p_{k+1}$ and $r_{f,k} = 0$. The corrected second input then equals $2Ex_k - w_k$, giving the exact ADMM iteration without a proximal term. Conversely, an exact first solve admits this lift, so the exact ADMM step has $E_k = 0$.

For $\rho > 0$, the graph pair is independent of the outer state and depends only on (x, z, a, b). Reusing this quadruple before the first check of the next outer iteration therefore reproduces the previously accepted graph pair. When $\rho = 0$, the graph pair depends only on (x, a, ℓ), since the exact second solve (31) is independent of w_k. Reusing this triple, including the same lift ℓ, likewise reproduces the previous graph pair. This is the warm start of Lemma 4.1. The outer results apply directly, and the inner-work results apply whenever the inner procedure satisfies their contraction assumptions for E_k^2. For $\rho > 0$, both solvers' elementary steps are included in the work count; for $\rho = 0$, it counts refinement steps for the first solve and its lift, with exact second solves required at the checks.

Finally, for $\rho > 0$, termination gives

$$\|d_k\|^2 + \rho^{-1}\|q_k\|^2 \leq \delta^2.$$

Therefore, $\|Ex_k - z_{k+1}\| = \|d_k\| \leq \delta$. Since $a_k + E^\top b_k = c(E^\top d_k + q_k)$, an application of the Cauchy–Schwarz inequality yields

$$\begin{aligned}\|a_k + E^\top b_k\| &\leq c\left(\|E\|\|d_k\| + \|q_k\|\right) \\ &\leq c(\|E\|\delta + \sqrt{\rho\delta^2}) \\ &\leq c\delta\left(\|E\| + \sqrt{\rho}\right).\end{aligned}$$

When $\rho = 0$, termination gives $\|d_k\| \leq \delta$ and the exact second solve gives $q_k = 0$, so

$$\|a_k + E^\top b_k\| = c\|E^\top d_k\| \leq c\delta\|E\|.$$

Together with $a_k \in \partial f(x_k)$ and $b_k \in \partial g(z_{k+1})$, these are the claimed KKT residual bounds for every $\rho \geq 0$. □

G Proof of the Davis–Yin Specialization

Proof of Proposition 5.4. We prove each item separately.

1. Clearly, the linear map C^* is surjective. Expanding $0 \in A(x_1, x_2, v)$ row by row gives

$$0 \in A(x_1, x_2, v) \iff \begin{cases} v - x_2 \in \tau A_1 x_1, \\ x_1 - v - \tau A_3 x_1 \in \tau A_2 x_2, \\ \theta(x_1 - x_2) = 0. \end{cases}$$

   Adding the first two inclusions and using the fact that $\theta > 0$ gives $-\tau A_3 x \in \tau A_1 x + \tau A_2 x$, which is precisely the desired inclusion. Conversely, if x solves (42), there exists $a_1 \in A_1 x$ with $-a_1 - A_3 x \in A_2 x$. Setting $v = x + \tau a_1$ then solves the inclusion problem.

2. Expanding $C\zeta \in A(x_1, x_2, v)$ row by row,

$$\begin{aligned} \zeta - x_2 + v &\in \tau A_1 x_1, \\ -\zeta - \tau A_3 x_1 + x_1 - v &\in \tau A_2 x_2, \\ \zeta &= \theta(x_1 - x_2). \end{aligned}$$

Set $\tau u := \zeta - x_2 + v$, so that $u \in A_1 x_1$ and, by the third row, $v = \tau u - \zeta + x_2 = \tau u - \theta x_1 + (1+\theta) x_2$. Substituting into the second row gives $x_1 - \tau u - \tau A_3 x_1 - x_2 \in \tau A_2 x_2$. That is, $x_2 = J_{\tau A_2}(x_1 - \tau u - \tau A_3 x_1)$. Substituting (44) into the three rows shows the converse statement. Then $C^*(x_1, x_2, v) = x_1 - x_2 + v = (1-\theta)x_1 + \theta x_2 + \tau u$, and adding $\zeta = \theta(x_1 - x_2)$ gives $\zeta + C^*(x_1, x_2, v) = x_1 + \tau u$. With $\gamma = 1$ and $(x_1, x_2, u) = (x_k, y_k, u_k)$, the terminal test $\|\zeta\| \le \delta$ of Algorithm 1 is line 5, its relative error test is line 7, and its update $w_{k+1} = w_k - \zeta$ is line 8. Line 4 computes the remaining components of the graph pair from (x_k, u_k), so improving (x_k, u_k) and recomputing y_k is the improvement step of Algorithm 1, and the residual $x_k + \tau u_k - w_k$ depends only on the output of the A_1-solver. Since (44) is a function of (x_1, u), starting the A_1-solver at the previously accepted (x_{k-1}, u_{k-1}) reproduces the previously accepted graph pair.

3. Let $(C^*(x_1, x_2, v), \zeta)$ and $(C^*(x_1', x_2', v'), \zeta')$ be two graph pairs, with u, u' as in item 2, and let $a_2 := \frac{1}{\tau}(x_1 - \tau u - \tau A_3 x_1 - x_2) \in A_2 x_2$, and similarly a_2'. Write d_i for the differences of the components, $\bar{d} := d_1 - d_2$, and Δu, Δa_2, $\Delta A_3 := A_3 x_1 - A_3 x_1'$ for the differences of the operator values. From (44), $d_3 = \tau \Delta u - \theta \bar{d} + d_2$ and $\zeta - \zeta' = \theta \bar{d}$, so

$$\left\langle C^*(d_1, d_2, d_3), \zeta - \zeta' \right\rangle = \theta \left\langle \bar{d} + d_3, \bar{d} \right\rangle = \theta(1-\theta)\|\bar{d}\|^2 + \theta\tau \left\langle \Delta u, \bar{d} \right\rangle + \theta \left\langle d_2, \bar{d} \right\rangle.$$

By the definition of a_2, $\bar{d} = \tau \Delta u + \tau \Delta A_3 + \tau \Delta a_2$, hence $\left\langle d_2, \bar{d} \right\rangle = \tau \langle d_2, \Delta u \rangle + \tau \langle d_2, \Delta A_3 \rangle + \tau \langle d_2, \Delta a_2 \rangle$, and therefore

$$\left\langle C^*(d_1, d_2, d_3), \zeta - \zeta' \right\rangle = \theta(1-\theta)\|\bar{d}\|^2 + \theta\tau \left(\langle \Delta u, d_1 \rangle + \langle \Delta a_2, d_2 \rangle + \langle \Delta A_3, d_2 \rangle \right).$$

The first two inner products are nonnegative by monotonicity of A_1 and A_2. By cocoercivity and $d_2 = d_1 - \bar{d}$, $\langle \Delta A_3, d_2 \rangle \ge L_3^{-1} \|\Delta A_3\|^2 - \|\Delta A_3\| \|\bar{d}\|$. Using $1 - \theta \ge \frac{\tau L_3}{4}$,

$$\begin{aligned} \left\langle C^*(d_1, d_2, d_3), \zeta - \zeta' \right\rangle &\ge \theta\tau \left(\frac{L_3}{4} \|\bar{d}\|^2 - \|\Delta A_3\| \|\bar{d}\| + \frac{1}{L_3} \|\Delta A_3\|^2 \right) \\ &= \theta\tau \left(\frac{\sqrt{L_3}}{2} \|\bar{d}\| - \frac{\|\Delta A_3\|}{\sqrt{L_3}} \right)^2 \\ &\ge 0, \end{aligned}$$

so B is monotone. For maximality, fix w and set $x_1 = J_{\tau A_1}(w)$, $\tau u = w - x_1 \in \tau A_1 x_1$. By item 2, this yields a graph pair with $\zeta + C^*(x_1, x_2, v) = w$, so $\mathcal{R}(I + B) = \mathcal{H}$ and B is maximal monotone by Minty's theorem [Bauschke and Combettes, 2017, Theorem 21.1]. The same graph pair gives $J(w) = C^*(x_1, x_2, v) = w - \zeta = w - \theta(x_1 - x_2)$ with $x_2 = J_{\tau A_2}(x_1 - \tau u - \tau A_3 x_1) = J_{\tau A_2}(2x_1 - w - \tau A_3 x_1)$.

4. The definition of y_k in line 4 of Algorithm 4 means $x_k - \tau u_k - \tau A_3 x_k - y_k \in \tau A_2 y_k$, i.e., $v_k \in A_2 y_k$, and then $u_k + v_k + A_3 x_k = \frac{1}{\tau}(x_k - y_k)$. The bounds follow from line 5 of Algorithm 4.

□